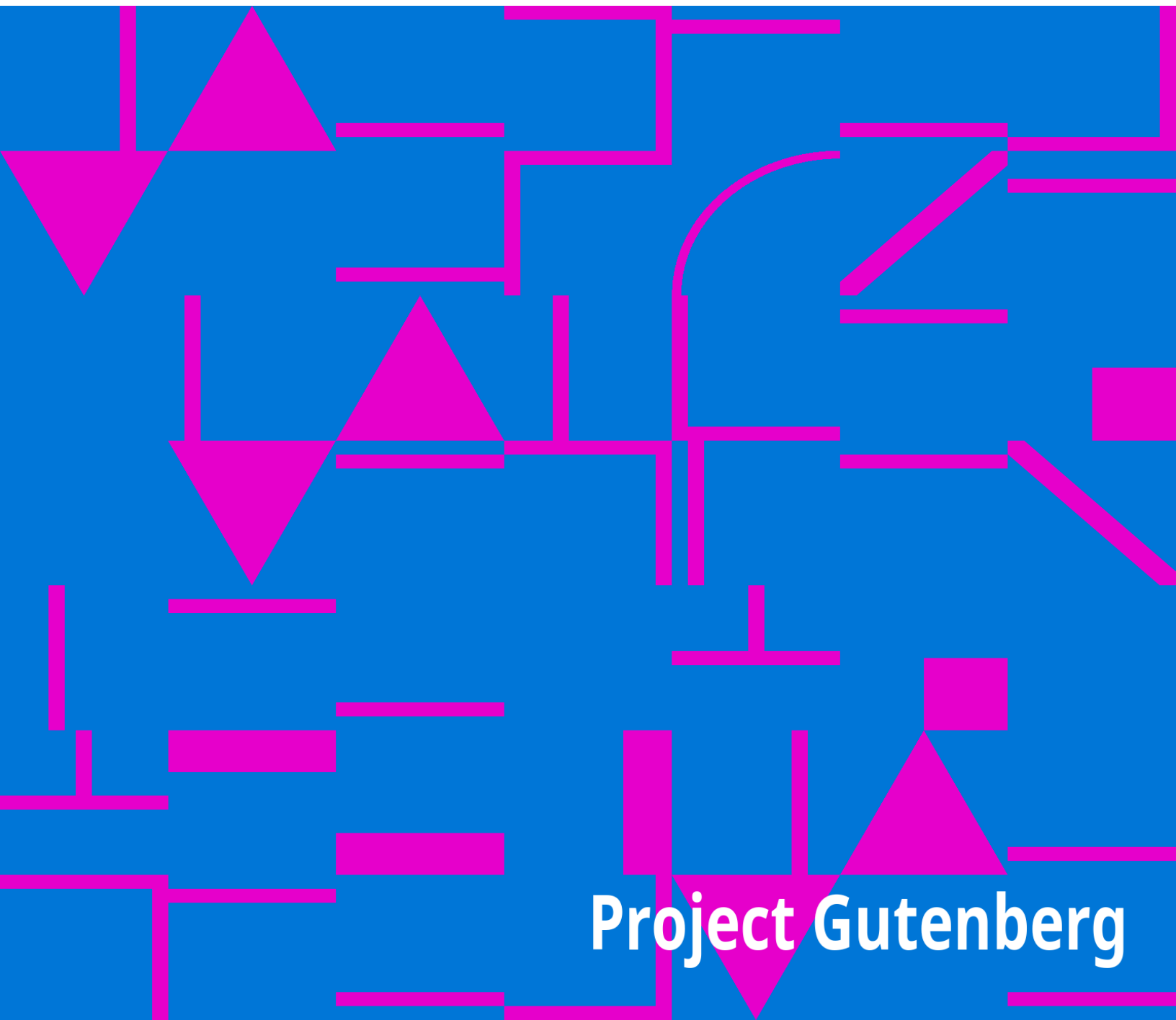


Harper's Round Table, September 1, 1896

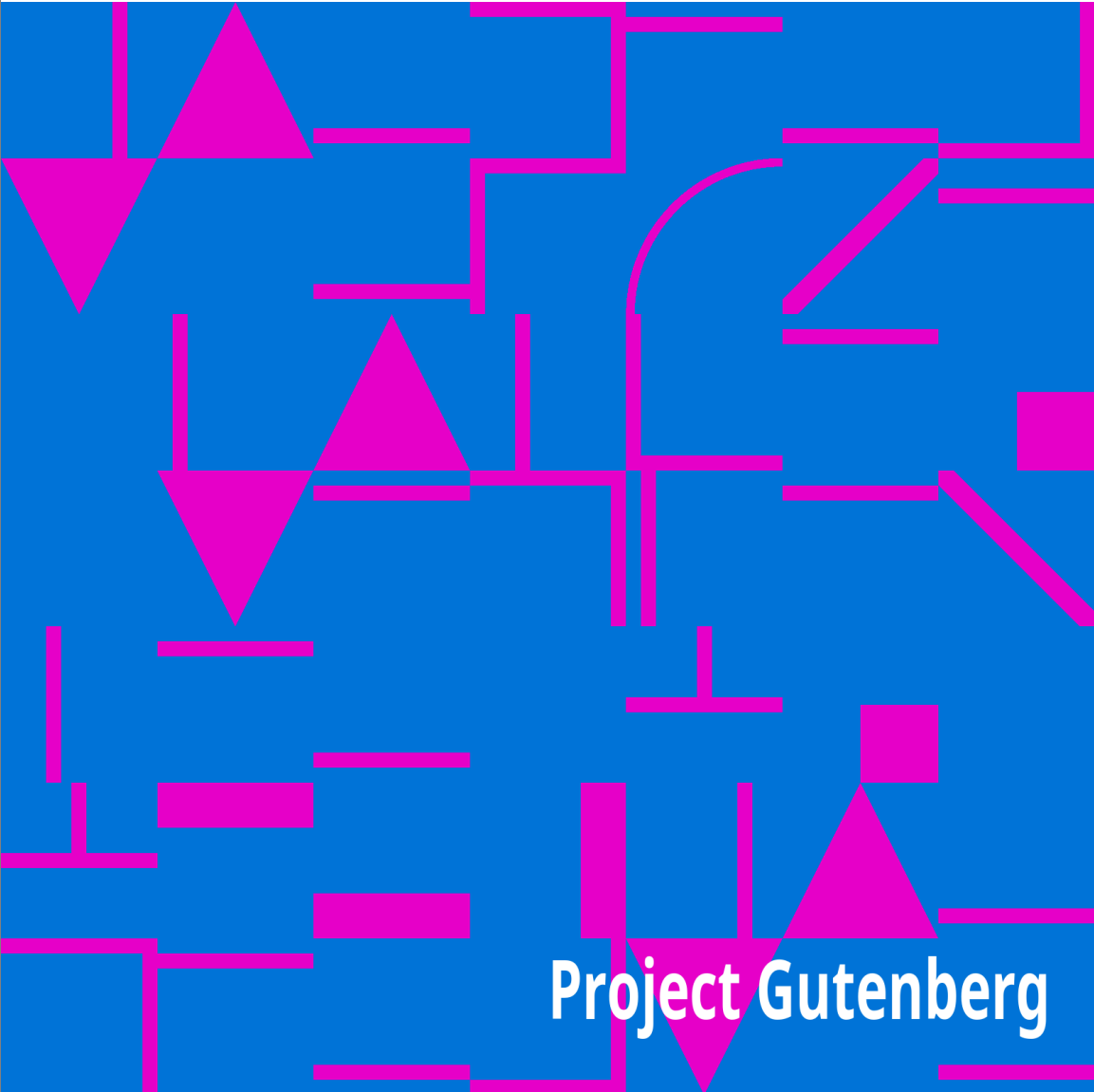
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Harper's Round Table, September 1, 1896

Various

An abstract geometric pattern on a blue background, featuring various magenta shapes including triangles, squares, rectangles, and lines, some of which are interconnected to form a complex, maze-like structure. The pattern is dense and covers the lower two-thirds of the page.

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IN THE OLD HERRICK HOUSE.

BY ELLEN DOUGLAS DELAND.

CHAPTER I.



here are but few dwelling-houses left which are occupied as such in that part of Philadelphia which was once so fashionable, the neighborhood of Independence Square. The rooms within the stately old mansions are now used by lawyers and other professional men for their offices, and business signs adorn the brick fronts without. There are one or two exceptions to this rule, however, and there was one house where the descendants of a long line of ancestors still lived in the home of their fathers.

The two Misses Herrick prided themselves upon having been born and brought up in the old house in Fourth Street, the same house in which General Washington had so often supped with their great-grandfather, when the table was adorned with the blue India oyster-dish and the egg-shell teacups, which were now kept behind the locked glass doors of the corner cabinet in the dining-room.

In one of the windows of this old house, in a room which fronted on the street, sat Elizabeth, looking out on the autumn rain, which dripped dismally through the leafless trees and flooded the brick pavement. Elizabeth was a niece of the Misses Herrick, and she had lived with them all her life.

It seemed a very long time, as she looked back upon it, although it was really only eleven years; but that is a great space of time when one is waiting; and Elizabeth had been waiting, ever since she had been old enough to know that she had a father, for his return.

He went away from America, she had been told, when her mother died, and that was when Elizabeth was a baby. Valentine, her brother, was almost three years older, and he had been sent to their mother's family in Virginia. The brother and sister had met but once or twice; for the aunts in Fourth Street did not like boys, and therefore did not encourage his coming there, and as Elizabeth had never been allowed to visit her Southern relatives, they were practically strangers to one another.

The Misses Herrick always spoke of the children's father as "poor Edward" when they mentioned him at all. This was when an infrequent letter arrived bearing a foreign post-mark. Elizabeth did not know why he should be poor, for his sisters were certainly very wealthy, and she had an indistinct idea, suggested to her by her old nurse, who was now dead, that some day she herself would have a great deal of money.

But that made no particular impression on her mind beyond the fact that if she did own money, she would like to give it all to her father if he were poor. Perhaps that was the reason he did not come home, because he could not pay for his passage.

Elizabeth thought it all out as she sat at the window this rainy afternoon, and she determined to question her aunts on the subject at the earliest opportunity. Julius Cæsar sat opposite to her, also looking at the rain. When a gust of wind rattled the window and swirled the dead leaves on the

pavement he gazed out more intently still; for although he was no longer young, and was extremely dignified, he was not above playing occasionally with anything so fascinating as a moving leaf.

The little girl would have led a lonely existence had it not been for Julius Cæsar, the cat.

The trouble was that Aunt Caroline was so occupied with her social duties, and Aunt Rebecca with the many lectures, concerts, and German or French classes which she attended, that there was little chance to speak with them.

Elizabeth did not see them very often, either—only at luncheon, or when she went out with Miss Herrick to be fitted for her fall or spring outfit, or after an altercation with Miss Rice, the governess.

On these latter occasions, which, it must be confessed, were very frequent, Miss Herrick was called in to act as mediator or judge, and Elizabeth found that she invariably took the part of Miss Rice in the discussion.

It was while she was thinking thus that her aunt Rebecca entered the room. Miss Rebecca Herrick was still a young-looking woman, tall and slender, and always beautifully dressed, and she was rarely seen without a book of some sort, for her tastes were distinctly literary.

When she came into the room this afternoon her face wore a preoccupied expression, and she was in evident haste. She did not see her niece sitting in the deep recess formed by the heavy curtain at the window until Elizabeth spoke,

"Aunt Rebecca, is it true that my father is poor?"

"Mercy, child, I did not see you there! How you startled me. What did you say?"

"Is my father a poor man?"

"Elizabeth, how absurd! Poor? Why should he be?"

"Because Aunt Caroline always says, when a letter comes from him, 'Here is a letter from poor Edward.'"

"Nonsense, Elizabeth! What ridiculous fancies you have! But don't stop me now with your questions. I am looking for the French book I am reading with Madame La Pierre. Have you seen it?"

"No," replied Elizabeth, not offering to look for it. "I am going to write to my father, Aunt Rebecca."

But her aunt, having found the book, had left the room.

"I am going to write to him, Julius," she repeated, stroking the cat's glistening white breast. "I do wish you were a fairy cat and could speak. It would be so nice to have some one to talk things over with. Never mind. When my father comes home, as he surely will when he gets my letter, I can talk everything over with him. Won't it be lovely, Julius?"

Then she left the window where she had been sitting so long and went to the writing-desk—her aunt Caroline's desk, so carefully arranged, with its silver implements and monogrammed paper. She would write to her father, as she had said, though she did not know how to address it. But fortunately her father's last letter, which had come that very morning, was lying open on the desk, with the date and the name of the place at which he was stopping written across the top.

She took an envelope and carefully copied the name, addressing it to Mr. Edward Herrick, and then she wrote the following letter—wrote it hastily, and without stopping to think what she should say:

"MY DEAR FATHER,—I hope you are well, and that you are coming home soon. I do not know why you stay away from us so long, unless it is because you have not enough money to come home with. Aunt Caroline always calls you poor Edward, so I think that must be the reason. I want to tell you that we are not poor here at all. Aunt Caroline and Aunt Rebecca both have lots and lots of money, and I have an allowance of seventy-five cents a week to spend as I like, only I have to buy my hair-ribbons out of it, because Aunt Caroline thinks I lose so many, and it is going to make me take better care of them if I have to pay for them myself; but it does not make a bit of difference, for they will get lost.

"I do not suppose that seventy-five cents a week will help you much to get home, but I am going to tell you something else. My old nurse Mary Ann, that died, told me once that when I was grown up I would have lots of money; she said I was an airess. I do not think that is the way to spell that word, but I will look it out in the dickshunary before I send the letter. I do not want you to think that I do not know how to spell, father dear. I read a book about an airess the other day, and it said she had a great deal of money, but she could not use it until she was very, very old—twenty-one, I think.

"Now, father dear, I have a suggestshun to make. Could not you borrow some money of somebody to come home with, and tell them you will pay it back in ten years? I have counted it up, and it will be ten years before I am twenty-one. It is a very long time, I know, but perhaps there is somebody who knows you well and will trust you. You can tell them that you know your daughter will pay it back.

"It seems strange that my aunts do not give it to you, for they have a great deal, I think; but I do not like to ask them to. They are very queer sometimes, father dear, though I do not like to say anything against your sisters. But won't you come home to me soon? I want you so much. We could live together, and my brother Valentine could come home too, and we should be so happy. I have thought it over ever so many times, and I think it would be too perfect. I really need you, father, and I will try to be just as much like my mother as I can possibly be. They say I look like her, for I have dark eyes and light hair; but I am not pretty, and she was. Aunt Caroline says I have an unfortunate temper. The words pop right out so fast when I get mad that I can't stop them, and so many things make me mad.

"But do come home, father dear. I need you so much; and please do as I say about the money. Come soon to your very loving and lonely little daughter,

"ELIZABETH

HERRICK.

"P.S.—The nicest thing in this house is Julius Cæsar. He is a cat, very large and black, with a white breast, four white paws, and one white spot in the middle of his back.

"E. H."

This was a very long letter, and the unformed childish hand in which it was written covered several sheets of Miss Herrick's best note-paper. When it was finished Elizabeth folded it and placed it in the envelope, forgetting to correct the misspelled words. She found a five-cent stamp in her aunt's

well-filled box—she had seen Miss Herrick put that kind of a stamp on her letters to "poor Edward"—and then going into the hall, she took an umbrella from the rack and sallied forth into the rain to mail the precious missive.

Elizabeth was mistaken when she told her father that she was not pretty. Her large dark eyes and the hair which hung over her shoulders like a mass of spun gold formed a striking contrast, but her cheeks were thin and somewhat pale, and her expression was too old for that of a child of twelve. Her lonely life was reflected in her face.

Her aunts did not intend to neglect her, but they were busy women whose own special interests came first in importance, and they did not understand the child. They thought that to feed and clothe her and to give her a beautiful home to live in was all that was necessary, in addition to the education, of which Miss Rice had charge.

They wearied of Elizabeth's questions, the result of long trains of thought carried on by the alert inquiring mind, and either refused to answer them or referred her to Miss Rice. The governess was one who considered it more important to know exactly how far it was in miles from the meridian of Greenwich to that of Washington, and what was the date of the eleventh battle of the Thirty Years' War, than to plunge into the subjects which interested Elizabeth.

Soon after the little girl's return from her expedition to the lamp-post Miss Herrick came in. She was in out-door dress, and she carried a card-case in her hand. Although she was rather below medium height, Miss Herrick's manner of holding her well-shaped head was so stately that she gave one the impression of being taller. Her features were regular, and there was not a trace of silver in the smooth dark hair which was never out of place. The Herricks were all noted for their beauty, and although Miss Caroline was well over fifty, was still a handsome woman.

"Are you there, Elizabeth?" she asked, in her evenly modulated voice; "it is a frightful day to go out, but I promised faithfully to go to Mrs. Ford's tea. Tell me when the carriage comes to the door. My umbrella seems to be wet. It is very strange. And who has been at my desk? The pen is still filled with ink, and carelessly flung down on the clean fresh blotter! Do you know anything about it, Elizabeth?"

"Yes, Aunt Caroline. I have been writing a letter there," said a small but courageous voice from the window-seat.

"Writing a letter at my desk? May I ask why and to whom? Does not the desk in your room afford opportunities for your correspondence?"

"It was easier to write it here. My room is so far off, and I wanted to get it done quickly."

"What nonsense! And to whom did you write?"

"My father."

"Your father! Elizabeth, how dared you, without my permission? Poor Edward! What will he say?"

"There you go again, Aunt Caroline," said Elizabeth, coming down the long room and standing at her aunt's side. "Won't you please tell me why you always call my father poor? Is he really and truly poor? Hasn't he any money? Do you suppose he is ever—really—hungry, like the Brady family in the back street?"

She asked these questions slowly and fearfully, and a solemn look came into the large brown eyes fixed so intently on her aunt's face.

"What nonsense you talk, Elizabeth! Tell me at once what you wrote to him about."

"Not unless you tell me why you call him poor," returned the little girl, firmly. "I ought to know something about my own father, I think, and you ought to tell me."

"You are extremely disrespectful. Leave the room at once."

"Very well; I will, Aunt Caroline. Only I think you might tell me, and I mean to find out somehow about my father. And I was the one who used your umbrella. I went out to mail my letter. And I used your stamp, which I will pay you back. And I got my feet soaking wet, and I don't mean to change my shoes. So there!"

"What a child she is!" thought Miss Herrick, as her niece disappeared behind the portière. "I wonder what she has written to poor Edward? What will he say? I trust he may never receive it. And she said her feet were wet. That will not do, for she will surely have a sore throat." And she touched the bell.

"James," she said to the man who appeared, "tell Marie to find Miss Elizabeth and change her shoes. Her feet are wet."

But it was some time before James told Marie, and still longer before Marie went in search of Elizabeth, and when she did the child was not to be found.

The house in Fourth Street was very large, with huge rooms and lofty ceilings, for it had been built in the generous fashion dear to our grandfathers. The drawing-rooms occupied the first floor, the sleeping-rooms of the two Misses Herrick the second, while Elizabeth had for her own one in the third story.

Then there were the "back buildings," on the first floor of which were the kitchens, above these the dining-room and library, and still farther above a number of rooms which were used for various purposes, such as the storing of furniture, camphor-chests, and the like.

There was one room in this part of the house which, to Elizabeth's knowledge, had never been opened, and, strange to say, it was fastened by a padlock. Elizabeth often wandered over the house when she had nothing else that was particularly interesting to do, and this padlocked door always possessed a strong fascination for her.

Why was it locked at all, and why was there a padlock on it? Was not an ordinary lock enough? There must be something very precious in there. What *could* it be? The mystery piqued Elizabeth's curiosity immensely. If she could only see behind that closed door!

On this rainy afternoon when she had been dismissed so summarily from the drawing-room she mounted the long double flight of stairs toward her own room. When she reached the third-story landing, however, her glance fell upon the locked door, which directly faced her.

"What is in that room?" she said to herself. "I must find out. Aunt Caroline won't tell me about my father, so I am going to discover things for myself. There is the front door shutting, so she is off, and Aunt Rebecca is taking her French lesson in the library. No one will hear me."

She turned and hurried down the thickly carpeted stairs, her flying feet making not a sound, and ran along the hall to her aunt Caroline's room. The lofty four-post bedstead, which had been made especially large for great-grandfather Herrick's famous height, seemed but of ordinary size in the great chamber, and the massive wardrobe and old-fashioned chests of drawers consumed but little of the space.

Elizabeth paused in the middle of the room and looked about her. If she could only see the key-bag which she knew Miss Herrick kept in her room. She would not like to open any drawers to find it.

It did not seem quite the right thing to go to people's bureau drawers. Fortunately it was not necessary. The key-bag hung on a rack near the dressing-table.

Elizabeth took it carefully down and ran up stairs again. Slowly and laboriously she tried each key to the little padlock. Not one of them would fit. There were thirty keys at least, and yet not one would open the door. What should she do? Disappointment only made her more anxious than ever to succeed.

Very dejectedly she returned to Miss Herrick's room and hung the bag where she had found it. She was turning away when she chanced to see a small Chinese cabinet of drawers on the dressing-table. It was curiously inlaid, and the corners were bound with silver, and it consisted of but two little drawers, the whole standing not more than four inches high. Elizabeth had noticed it before on the table when she had been in her aunt's room, and she had always admired it.

She took it up and looked at it. One of the little drawers slipped out as she held it, and within lay two keys, one large and the other small, and they were tied together with a ribbon. With a half-suppressed "Oh!" of delight she seized them and ran up stairs.

One key fitted the lock of the door, the other the padlock. With perfect ease she turned them and entered the room.

[TO BE CONTINUED.]

THE SAILING AND RACING OF THE SMALL BOAT.

BY DUDLEY D. F. PARKER.

A great deal of real solid pleasure may be had from the handling and racing of the smaller classes of sailing-craft.



THE "TUCK-UP."

The boat possessing all the peculiarities of the small boat to the greatest extent is the "tuck-up," a type seemingly a cross between the ordinary cat and a "Whitehall" row-boat, and having an extremely easy under-water body. These little boats range from sixteen to nineteen feet, and have a very moderate beam as compared to the cat-boat, and are rather deeper. The entrance is sharp, and the stern lines are much cut away, making a hull that can be driven through the water at a high speed, though lacking in stability. It is frequently the case that the spars and sails of these boats are so large that they are top-heavy from this alone, and often the boat has to be run alongside a dock and a man hold it right side up whilst the crew clamber aboard. Ropes are frequently fastened at intervals to the centreboard trunk to assist the men in hanging on when the boat is away over. In the sketch the crew is represented as laying flat on the deck for windward work.

They are essentially a "racing-machine," their speed being developed by a large sail, big centreboard, and all the men and bags available to keep the whole right side up. The rig is the regular cat, and everything is very strong. On account of the small displacement of hull and the large sail and ballast, the motions of the boat are much intensified. A squall striking the large sail area will throw her down in an instant, and as soon as it is over she rights as quickly. It will be readily seen from the foregoing that the crew should watch the boat intently, and be prepared to hang out over the side or scramble inboard, as the occasion may require. The problems presented for racing these boats are about the same as those of the cat-boat (see article in HARPER'S ROUND TABLE No. 827), but there are several things that should always be kept in mind. These boats are quick in stays, obedient to the helm, and change course rapidly when gybed. On account of comparative lack of beam, the deck at the sides is narrow and the coaming is low, so as to permit the men to lay over the side, and in consequence the danger of taking water aboard is greatly increased. In fact, it is a frequent sight in a race to see a man busily engaged in bailing out the water that has dashed in over the bow or come in over the lee gunwale. The quick manner of heeling of these boats greatly increases the chances of getting the sail in the water, and you should keep a sharp lookout for this, as it is liable to result in an upset. At the same time you do not want to let the wind out of the sail, or the weight of the men on the upper side will dump the boat to windward. The principal duty of the crew will be to act as shifting ballast, and the greater the rapidity with which the motions of this important duty are performed, the more the boat's speed will be helped. When the boat starts to heel, it should be met quickly, so as to prevent her getting away over and wallowing through the seas, the men lying out or sliding in quickly, as the wind's force varies. The duties for each man are about

the same as in the cat-boat, having a man at the tiller (captain), sheet-tender, centreboard and halyard man, and if the day is at all windy a light man had better be assigned to bail the boat when necessary.

St. Lawrence skiff-racing is not general, though in some parts of the country it is indulged in. It will be sufficient, perhaps, to pass over it with a few words. The boat is in many respects a large canoe, and hence depends entirely on the crew to hold it up. The rigs employed are enlarged canoe rigs, *i.e.*, two fore-and-aft sails of some character—"bats' wings," "Mohicans," etc., etc.

The sternmost sail, or "jigger" (sometimes called "dandy"), has a tendency to throw the boat's head up into the wind, and as there is no counteracting influence of a jib, these rigs sail very close to the wind.

In going about, the jigger is a great aid, and should be hauled in flat when rounding up, and trimmed properly again when on the other tack. In gybing, the jigger is more of a hinderance than an aid, as in making a gybe it is necessary for the boat's head to fall off the wind. To make a neat job, the wind pressure in the jigger should be reduced as much as possible as the mainsail is coming inboard. In running before the wind the sails should be placed "wing-an'-wing," that is, on opposite sides of the boat, so as to get full benefit of the wind and ease the steering. When running this way you should watch the boat carefully, lest some small change in wind or course would cause one of the sails to gybe over.



**THE ST. LAWRENCE
BOAT.**



RACING BATEAU.

The racing small boat pre-eminent of this section of the coast is the "bateau"—a half-round-bottom type possessing some of the qualities of the flat-bottomed row-boat and the sea-skiff. They are usually in the neighborhood of eighteen feet, and rather narrow, with a sharp bow and long tapering stern-lines. The bottom board, or "keel," is about eighteen inches wide in the centre, tapering to a point at each end. From it the sides are built up out of two or three wide planks riveted together. The stern is really an overhang, but has a skag built on underneath, and terminates in a perpendicular stern-board. The rig generally employed on these boats is the "skiff rig,"

though occasionally the mainsail with gaff and halyards is used. The use of a jib presents many new problems in sailing, and will necessitate some thought and study on the part of a boy whose experience has been confined to the handling of a cat-boat. In the jib-and-mainsail rig there are two opposite forces at work. The mainsail, as in the cat-boat, throws the boat's head in the wind, only this is more pronounced, as the mast is stepped further aft. This can readily be seen.

Let us suppose our boat to be a lever pivoted on a fixed point, and free to swing on this point. The point in the boat that takes the place of the pivot of the lever is the centre of lateral resistance (resistance to sliding sideways). This centre will be somewhere amidships, and it is clear that whichever side of it the greatest wind pressure is exerted on the sails, that part of the boat will

have a tendency to drift with the wind, so swinging the other end around. In the mainsail the greatest power is aft of this centre, hence the stern falls off and the bow goes up into the wind. In the jib the reverse is true. The power here is applied forward of the centre, and hence the bow falls off. These two neutralizing forces should be borne in mind when handling a jib and mainsail, and made judicious use of.

Moderate observation and some thought of this principle will enable a boy to analyze the behavior of his boat, and to trim his sails so as to correct errors and get their full benefit. The jib may be trimmed so that the boat will almost steer itself, though the sail should not be in so flat that the mainsail cannot cause the boat to luff up when you let go the tiller. When beating to windward or close-hauled the pressure on the head-sail must be lessened, and so the jib should be given considerably more sheet than the mainsail. When you wish to go about, the rudder and mainsail are handled in the same manner as in a cat-boat. (See HARPER'S ROUND TABLE No. 827.) After letting go of the tiller and starting to haul in the main-sheet, and when you notice the boat is rounding up, let slack the jib-sheets, and when the boat is on the other tack, trim it in tightly, so as to carry the boat's head over quickly, and then trim properly after the mainsail fills. In gybing the jib is of much assistance. The main-sheet and tiller are handled about the same manner as in the cat, but perhaps another

caution should be given about not forgetting to haul the centreboard up. Never fail to see that the board is up before throwing the boat off with the rudder. When about to put the rudder over, after pulling the board up, trim the jib in flat, and it will aid greatly in swinging the bow off; at the same time, if kept trimmed in until the boat's course is settled, will check to a great extent the tendency of the boat to run up into the wind after the sail goes over. When running off the wind the jib should be in nearly as flat as the mainsail, but you should always remember not to have the jib in so tightly as to destroy the luffing power of the boat. If a sudden squall strikes the boat, let go the jib-sheet, and it will luff up quickly. When running before the wind the jib may be used as a spinnaker (see sketch of skiff) by placing the sails "wing-an'-wing," and if there is not enough wind to hold the jib out, a light pole or an oar may be used for that purpose. If the racing rules permit, a pole may be set over the side abreast of the mast, and the jib rigged as a regular spinnaker.

The crew should be disposed, when possible, so that the boat will set deeper in the stern than in the bow; that is, the bottom board at the bow should be about two or three inches under water, so that the cut-water will part the waves. If the bow is lifted out so as to bring the flat bottom board in contact with the waves the boat will "smash" its way through, and each wave will stop the headway. If the bow is too deeply in, a larger sea than usual is liable to come aboard, and in addition the rudder will be lifted out to a certain extent, and the boat will steer badly, on account of lack of rudder area and the increased lateral resistance of the bow.

Sneak-boat racing is also popular in some sections. The sneak-boat is very much like the cat-boat, and is about as uncapsizable as a boat can be. Its lines are all full, mostly arcs of circles, the sides being "flaring." The long spoon-shaped bow and broad full stern overhang give great stability and displacement when heeled over. A fore-and-aft sail of some character is used, and the boats are



SEA-SKIFF.



**RACING SNEAK-
BOAT.**

handled much like a cat-boat. Under favorable conditions they develop good speed. They are unsurpassed as a boat for the beginner in sailing or racing.

Sea-skiff racing is indulged in by the fishermen, and these speedy boats afford some good sport.

But to leave these special types, and coming to small-boat racing in general: What qualities are essential to the successful racing skipper? I should link two together as outweighing all others—good judgment and spirit. They seem to stand together; one without the other is liable to lead to unsucces. Judgment without the spirit to make best use of it will lead to over-caution. Spirit without judgment verges on rashness, and cannot but lead to disaster.

The racing-man must think quickly, and act immediately and decisively. He must train himself to take in situations at a glance and determine the policy he will pursue immediately.

The boat should be sailed with dash. Your main idea should be to get speed, and everything else should be subordinated. Injury to boat should not be considered, unless it is liable to cripple her permanently, and this risk is very often pardonable. No thought should be given to the crew's comfort; they should be regarded as machines, for the time being.

In short, the tuck-up, skiff, bateau, or small boat of any character should be sailed like a large toy boat. If a gust throws her down, get the crew out over the side if necessary, so that there is only enough inboard to keep them from falling off. Don't be afraid of taking water aboard; when there is enough in to be troublesome, bail it out.

A racing-man must be observant. He should notice where the tides are the strongest, and also which way the flow is at the time of the race. If possible, he should make himself familiar with all this before the race, and it should have some effect on the course of his boat. For instance, suppose one of the legs of the course takes you up a channel, as the outlet of a river or bay. If the tide is against you, you should hug the shore and avoid the deep water, as the tide always runs strongest in the centre. Now suppose you have rounded the mark, then the tide is in your favor, and it is clearly to your interest to get in the middle of the channel, so as to have its full benefit. It will easily be seen what might be lost by a skipper not knowing or observing which way the tide was flowing. I witnessed a very good example of this last year, in the Labor Day Races, in the Horseshoe, Lower Bay. The course was a thrash to windward out to the Sou'-west Spit and return, from a starting-line near the mouth of the Navesink River. There was a strong flood-tide and a light wind, causing the fleet to split into two sections, one tacking in under the Hook, and the other taking the straighter course for the mark. The latter got the full handicap of the incoming tide, and was left far behind the boats which had gone inshore to escape it.

At all times you must be on the lookout to take any little advantage an opportunity offers, and to interfere with an opponent's wind. If possible, when passing a boat, always go to windward, but do not enter into one of those senseless luffing matches, which will practically take the two boats engaged out of the race.

The price of success in racing is vigilance, observation, decision, and no set rule can be laid down for the racing captain. A great deal will depend on opportunities and the manner he makes use of them. There is some luck in boat-racing, but much that is ascribed to luck is due to the forethought

of the captain. Very often you hear that such and such a boat, with her usual good luck, received the first advantage of the changing breeze. How is it that in these cases it is usually the same boat; can we entirely and justly ascribe it to luck?

OLD HAMMER-THE-ROCKS.

BY WILLIAM DRYSDALE.

Down in Brooks County, in southern Georgia, people are still laughing over the great joke on Dick Weston. Dick is rather a favorite, too, though he is a Northerner; but people always laugh at a fellow who tries to play a sharp game and is tripped up.

"Now don't you be like Dick Weston," they tell their boys, "and waste your time waiting for dead-men's shoes. I want to see you like his cousin Larry—able to show that you amount to something."

There would not have been any joke on Dick Weston, and perhaps Larry Weston would not have made his great strike, if it had not been for the civil war. Major Weston, of Massachusetts, went down to Georgia in the Union army, and after the war he bought a big plantation in Brooks County, and made a heap of money. Then years afterward his brother Henry, a merchant in Boston, went down and bought half the plantation; and there the two lived, and still live, with every comfort in the world, except the comfort of good wives; for they are both old bachelors.

Of course it was common talk in Dick's family, and in Larry's family, too, that the two bachelor uncles were very rich, and that the two nephews would most likely be their heirs. Larry never paid any attention to this talk, for his head was full of other things. But it was very different with Dick.

"Your nose is out of joint, young man," Larry's father said one day, giving him a poke in the ribs; "I have a letter from your uncle George, and he says Dick is going to make his home with them."

"Why shouldn't he, if they want him?" Larry answered. "I'm sure I don't envy him. For my part, I'd rather make my own way than depend upon somebody else."

That was two years before Larry's visit to Georgia, and Dick put in two years of faithful work trying to make himself agreeable to his rich uncles. He was so sweet around the house that his old school-mates would hardly have known him. His uncles' slippers were always warmed before the fire on cold mornings; and whatever they liked, Dick liked. If they had said that cotton grew on chestnut-trees, Dick would have said so too.

Uncle George and Uncle Henry laughed in their sleeves at Dick's wonderful affection, for they had been in the world a good while and knew a thing or two. But they liked him, nevertheless, for he had pleasant ways and was a handsome fellow; and the neighbors liked him, though they said, "he's playing for big stakes, and he'll likely win."

All this time Larry was attending strictly to his own business, and learning a heap about the rocks that lie in the ground. That was his hobby. Other things he learned because he knew they were necessary; but mineralogy he studied because he loved it. His ambition was to go out into the Southwest when he was old enough, and find gold or silver or some other valuable stuff where nobody suspected its existence. His room was full of cases of broken rock, and he rarely went out without his little hammer. It was a standing joke in the house that the police were looking for him for breaking corners off the curb-stones.

The two uncles evidently kept an eye upon Larry, for as soon as he was done with school they invited him to spend a month on the plantation. That was when he was just past eighteen.

"No, *sir!*" he said to his father. "I wouldn't go for a farm. There's no meaner business in the world than trying to curry favor with rich relations."

"But think what a chance you'd have to geologize, or metallurgize, or whatever you call it!" his father suggested. "There must be rocks in that country, and you could break them by the ton."

That put a new face on the matter. Larry had nicked specimens from the rocks of Harlem and Manhattanville and the Palisades, but the Georgia rocks would give him a new field; and for strictly professional reasons he decided to go.

As Larry had never seen a plantation before, he found everything very different, of course, from what he expected. It was much larger, to begin with—more than 4000 acres—and he could roam about all day without going off his uncles' land. One big cotton-field contained more than 2000 acres, and one morning he counted eighty-five men, women, and children at work on it hoeing cotton. At the same time twenty men, with mules, were cultivating corn on the other side of the place; and the rows of cabins for the workmen looked like a village. Still he was rather disappointed. He expected to see beautiful green lawns shaded by fine old oaks, and beds of brilliant flowers, and everything as smooth and clean as a rich man's place in the North. There were plenty of oaks and flowers, to be sure, but there was no skilful gardener to keep them trimmed smooth. The house was old, and as there was no mistress, the negro women had their own way, and everything was in disorder.

But the plantation was no more of a disappointment to Larry than Larry was to the plantation—at first. He was so quiet and thoughtful, not at all such a "hail-fellow-well-met" with everybody as Cousin Dick. There were horses to ride and birds to shoot and fish to catch, but he took no interest in such things. When he could have a bit of rare rock to examine, and some acids to test it with, he liked that better than the plantation sports.

"Come, Larry," Uncle Henry said one morning, "and take a gallop with us. We are just half-way between Quitman and Thomasville here, and we'll go whichever way you like."

"I don't know much about horses, Uncle Henry," Larry answered, "and don't want to break my neck. Besides, I have some queer specimens here that I want to test; so I guess I'll keep house while the rest of you go."

Dick was suspicious of his cousin, and always ready to make him the butt of his jokes. Before Larry had been on the plantation a week he had nicknamed him Post-Pliocene, Alluvium, Kill-Sport, and a dozen other things; but Old Hammer-the-Rocks was the favorite title, and it was so appropriate that it always made Larry smile. And Dick's love for his cousin did not increase at all when he found, after a few days, that the uncles had discovered that Larry was full of information about many things, and liked to talk with him on subjects that Dick knew nothing whatever about.

On the morning of the day when Dick unwittingly played the great joke on himself, Larry was sitting in the library just after breakfast, looking over the State Geologist's Report. Dick soon joined him, and burst out with:

"Come on, Old Hammer-the-Rocks! We're going after birds. Shake yourself up once and come along."

"I never killed a bird in my life, Dick," Larry answered, "and I'm not going to begin to-day."

"Ah, indeed!" Dick sneered. "Don't approve of such sport, I suppose."

"No, I can't say that I do," Larry replied, very good-naturedly. "It may be sport for you, but what must the birds think about it?"

Uncle George, the Major, bustled into the library after his cartridge-belt just in time to hear this conversation.

"Who's that don't approve of killing birds?" he asked, pretending to be very angry.

"I don't, Uncle George," Larry replied, "It's a heap more pleasure to hear them sing in the trees than it can be to kill and eat them."

As he spoke he could hardly keep from laughing outright at the expression on Dick's face. Dick seemed perfectly horror-stricken to hear one of his rich uncles reproved in such a fashion.

"Well, there *are* more important things in the world, that's a fact," the Major laughed. "You go ahead with your uncle Henry, Dick, and I'll join you later. I want to have a little talk with Larry."

"This is a nice state of affairs!" Dick said to himself, as he went out. "I never contradicted Uncle George in my life, but he never keeps *me* in the library for a private talk!" and he began to wonder whether a little independence, after all, could be better than his own way.

"You mustn't let us bore you with our country sports, Larry," the Major said, when they were alone. "We have nothing else to amuse ourselves with, but you have. I am glad to see you so much interested in geology and mineralogy; the knowledge may be useful to you some day. That is all I wanted to say; I want you to enjoy yourself while you are here, and enjoy yourself in your own way, whether it's shooting birds or hammering rocks. Now I'll go and murder a few birds, as I know nothing about rocks myself."

Left to himself for the whole morning, Larry determined to follow the little creek that ran southward, and see what its banks had to offer. Besides, he knew that by following it for three or four miles he would come to the Florida line, and he wanted to be able to say that he had been in Florida. So, with his hammer and his little black bag, he set out.

The Weston place, like most of the big plantations, has its own gin-house, where the cotton is passed through a machine that separates the fibre from the seed; and the gin-house is always built over a running stream, so that the water will turn the big wheel.

He did not imagine, as he followed the gin-house stream, how fate was arranging everything for him that morning. He had not gone a mile before the sky began to grow black.

"No matter," he said to himself; "if it should rain hard, there is the gin-house a mile further on. I can find shelter there."

The rain came in due course, a real Southern downpour, and he hurried along to the gin-house and went in.

It is a ghostly old place, that Weston gin-house, built of solid timbers many years ago. The floor is on a level with the ground, and a big double door lets in light for the machine. But beneath the floor is a deep gully washed out by the stream, dark and damp. From a trap in the floor, steps lead down into that black hole, where the big wheel turns, and a maze of great timbers support the wheel and the building.

Just as Larry stepped into the gin-house and closed the door, Dick emerged from the woods beyond, hurrying home out of the rain.

"There's Old Hammer-the-Rocks gone into the gin-house," he said to himself, "and I believe I'll get a little wetter for the sake of giving him a scare. Instead of going in myself, I'll put the prop against the door and fasten him in."

The rain made so much noise that there was little danger of his being heard, and he went boldly up and fastened the doors.

"Now he'll have a time of it," Dick said to himself. "The only way he can get out is to go down past the wheel, then climb down the timbers to the bank of the stream, and crawl under the siding and climb up the rocks. Those timbers are slippery, too; but if he breaks his neck that's his lookout, not mine."

Up to a certain point everything worked precisely as Dick had foreseen; but who in the world could have imagined what was to happen afterward?

When the rain let up a little Larry tried to open the doors, but they would not open. He pushed and pulled, but the heavy doors would not budge. Then he began to investigate. It was very dark inside, but through the trap he saw that down below the wheel there was more light. Though the house was weather-boarded all the way down, there was an open space at the bottom for the water to run through. That was the only way to get out.

He felt his way cautiously down the dark stairs to the platform by the wheel, expecting every minute to put his hand on a lizard or a centipede or some other unpleasant creature. The wilderness of thick timbers down there reminded him of some church belfries he had been in, but the belfry timbers were not so soft and slimy to the touch. From the side of the wheel he started to walk across a short timber to the wall, so that he could climb down to the bottom of the gully. But the timber was worse than he thought—half rotten, slippery with moss and slime. His foot slipped, and he fell, not into the shallow water, but upon the rocks by the side of the little stream.

That was just as Dick had more than half wished. Larry lay stunned upon the rocks beneath the old gin-house.

But then the great joke on Dick Weston that all of Brooks County is still laughing about began to develop.

Instead of lying there mangled and bleeding, Larry got up and found that he had no hurts beyond a few bruises. He was in a spooky place, but he forgot all about the fall and the mystery of the closed door when he saw that there were more rocks all around him than he had found before on the whole plantation. The stream had cut its way between walls of rock, and the ledge was littered with loose bowlders, large and small.

He picked up some small specimens to put into his black bag, and something in the feel of them startled him. It was a curious combination of roughness and smoothness that his fingers touched. He knew he had felt that species of rock before, but where? Certainly not often. It must be something uncommon. He picked up as many specimens as the little satchel would hold, and crawled out into the daylight.

He was twenty feet below the surface, between two walls of rock that extended as far up the gully as he could see. He touched one of the rocky walls, and felt again that curious sensation of roughness and smoothness.



**LARRY'S DISCOVERY UNDER
THE MILL.**

He chipped off a small piece with his hammer, and sat down on a big boulder to examine it.

"Now I have it!" he exclaimed. "It was in the Museum of Natural History that I saw and felt this stuff. But it can't be that this is a great deposit of—"

He was almost afraid to speak the word, for perhaps he was mistaken, after all. He took the hammer again and pounded part of his specimen into powder, felt it, studied of it, and tasted it with his tongue.

Ten minutes later he was hurrying across the wet fields towards the house, his pockets bulging with specimens broken from a dozen different places. When he reached home he went straight to his room, and soon filled the air with the unpleasant odor of acids poured upon pounded rock.

The Major and Uncle Henry and Dick were in the library when he went down stairs, talking over their morning's sport.

"Oh, you missed it, Hammer-the-Rocks!" Dick exclaimed. "We had a royal time."

"I had a pretty good time too," Larry answered. "I explored the cellar of the old gin-house, and found some very interesting specimens." And he unloaded his pockets and the satchel upon the library table.

"Specimens!" the Major exclaimed, picking up one of the pieces. "Why, this is just our common rock. I'm afraid you have fooled yourself this time, Larry. The whole place is underlaid with this stuff—more's the pity!"

"Is it?" Larry asked, very coolly; "that's good. What name do you give it?"

"Oh, we don't give it any particular name," the Major replied, tossing the specimen back to the table; "just ordinary rock."

"Then you won't mind my giving it a name, Uncle George," Larry went on. "I call it wavellite; it is worth about eight dollars a ton, just as it lies."

"What?" both the uncles exclaimed together, springing to their feet. "Eight dollars a ton!" And Cousin Dick began to look uncomfortable.

"Fully that; perhaps more," Larry continued. "I consider it a better find than a vein of gold, for it is safer. It is the most valuable phosphoric rock known to commerce, and has never been found anywhere but on one little island in the West Indies. Wavellite, or Redondo mineral, is the commercial name of it. But you must not depend solely on my opinion. Have the specimens examined by an expert."

The bird pie received little attention that day. Uncle Henry took an afternoon train to Savannah with half a pack of specimens, and returned two days later with the expert's verdict: "Wavellite beyond doubt." Soon acres of growing cotton were turned into big holes in the ground where the mining was done. The beauty of the plantation was spoiled by the heaps of rock thrown up, but its value was increased many times over.

It was only last week that the Major wrote to Larry's father:

"Of course you will not think of calling Larry home. He has charge of all our mining operations, with a ten-per-cent. interest in the output that will make him a rich man in two or three years. Dick, I am glad to say, is making himself useful too; he is Larry's clerk. I suppose we should never have known of the wealth under our feet if it had not been for Larry."

It was Dick's own fault that the story of the gin-house leaked out. He told one of his intimate friends about his "bad luck," and it was soon all over Brooks County. As the planters ride past and see him keeping tally in his little book, they often call to him:

"Hello, Dick! What will you take to fasten somebody in *my* gin-house?"

A VIRGINIA CAVALIER.

BY MOLLY ELLIOT SEAWELL.

CHAPTER XII.

As soon as George had spoken he disengaged himself gently from his mother's arms. She was still weeping, but blessing him.

"God will reward you, my son, for this yielding to your mother!" she cried.

"I don't know, mother, whether I deserve a reward, or not," he answered, in the same strange voice in which he had first spoken. "I am not sure whether I am doing right or not, but I know I could not do otherwise. I did not yield to your command, but to your entreaty. But let me go, mother." And before she could stop him he was out of the room, and she heard his quick step up the stairs and his door locked after him.

He tore off his uniform as if every shred of it burned him, put on his ordinary clothes, and then sitting down on the bed, gazed blankly before him.

And blank looked the life before him. He had suffered himself to dwell upon the thought of a naval or military career until it had become a part of his life. He foresaw that the same strange weakness on his mother's part which kept him from joining the navy might keep him out of the army. True, if there should be war between the French and English in the Northwest it would be his duty to defend his country, and no pleadings could keep him back then; but that was only a contingency. And, in any event, he could not again ask the help, in getting a commission, of the only persons who could serve him—his brother Laurence, and Lord Fairfax—after this unfortunate ending of his first attempt. And, worst of all, he was not sure that he was right, and he was very sure his mother was wrong. That of itself was a staggering blow. He had always fancied his mother perfect, and her weakness, her blind partiality for him over the rest of her children, at once shattered his ideal. She was a true and devoted mother, but in a great emergency she showed a tender unwisdom that seemed foreign to her character. George did not love her any the less for this, but he realized that after this he must think and act for himself. She had not thought of how far he was committed in the matter, or that his brother Laurence might be justly offended at his course—she only thought of the anguish of giving him up. It was all hard and inscrutable to the boy, sitting with rigid face and dry eyes, gazing before him and seeing nothing. He did not know how long he sat there. He heard Betty's light step, and lighter tap upon the door, and she called him softly through the key-hole.

"Go away, dear Betty," answered George; "I can't see anybody just now."

It seemed to him days, not hours, before he heard the bell for dinner. He gathered himself together and went down stairs. Betty almost cried out when she saw him, he was so haggard. His mother saw it too, and it made her heart ache; but in her heart she felt that it was better to have him as he was than to say good-by to him forever, which she was firmly persuaded would be the case had he gone in the navy. Madam Washington, being naturally a woman of great integrity, was not at ease in her mind. She had not forgotten the light in which she would appear before Laurence Washington and Lord Fairfax. She read again and again that letter from Joseph Ball, which George

had appalled her by calling both ignorant and foolish. She had been taught to think brother Joseph a monument of wisdom, but she was not so sure of it after having acted on his advice in this great event.

At dinner both George and his mother were perfectly composed and polite. Neither the children nor the servants knew that anything was the matter, until Betty betrayed it. But little Betty's heart was so full for George's disappointment that she could not eat her dinner, and tears dropped upon her plate. Towards the last of the dinner one of the little boys suddenly exclaimed,

"Brother, I saw you in your uniform this morning; are you going to wear it every day?"

At this Betty burst into a loud sob, and getting up from the table, rushed to George and threw her arms about him. George rose and led the weeping girl out of the room. Usually such an infraction of discipline and table manners as George and Betty leaving the table without permission would have been strictly prohibited. But their mother saw that these two young souls were wrought up to the keenest distress, and as she had gained her victory she could afford to be magnanimous.

"Betty," said George, hurriedly, when they got out of the room, "put on your hood, and let us go into the woods. It makes one feel better, when one is sad, to go into the woods."

The day was dull and overcast as the boy and girl, hand in hand, tramped across the fields to where the fringe of cedars formed the advance-guard of the woodlands. George held Betty's hand very tightly in his. *She* understood him, at least.

They said but little until they were well in the heart of the woods, and had sat down upon a fallen tree. Then George, laying his head on Betty's shoulder, burst into tears, and cried as if his heart would break.

No creature was ever better formed to feel for others than sweet little Betty. She had never seen George weep like that; but she was not frightened or disconcerted. She only laid her wet cheek against George's, and sighed so deeply that he knew that his burden lay as heavy on her heart as on his. Presently, when he had become more composed, Betty spoke:

"Brother, hard as it is, I am glad of one thing—nobody can say anything to you about it, after you have said that you gave way to our mother, for no boy, or man either, can let anybody in the world find fault with his mother."

"Yes, Betty," answered George, sadly. "I will not be such a poltroon as to let any one say my mother has not acted right."



"SHE MEANT TO ACT RIGHT," SAID BETTY.

"She meant to act right," said Betty; "but—" Betty paused, and the brother and sister looked into each other's eyes and said no more, but each understood the other.

"Of course," sighed Betty, "it would have been the hardest thing in the world to have you go away; but if you wanted to go, dear George, and it was best for you, *I* would have given you up; and I would have tried not to cry when you went away, and I would have thought of you every single day while you were away, and if you had not come home for ten years or twenty years, I would have loved you just as much as ever."

George had always loved Betty dearly, but he felt now, at the hour of his cruellest disappointment, what it was to have that tender sister, to whom he could reveal his whole heart. Much as he loved his brother Laurence, deeply as he revered Lord Fairfax, and with all his love and reverence for his mother, he felt obliged to keep up before them a manly fortitude; but Betty was young and inexperienced, like himself, and, because of that, in some ways she was nearer to him than anybody else.

The two sat there until late in the afternoon, and so quiet were they that a squirrel came boldly out of his hole and hopped past them, and a robin, with a weak little pretence of a song, in spite of the wintry weather, swung within reach of them. It was nearly sunset before they took their way homeward. George, like all boys, was not glib of tongue in expressing his emotions; but when they got to the edge of the woods he kissed her, and said:

"Betty, I don't know what I would have done if it hadn't been for you this miserable day."

The little sister's loyal heart grew almost happy at this.

A hard task remained for George. He had to write to his brother Laurence and to Lord Fairfax, announcing what he had done. They were not easy letters to write, but he carefully refrained from any hint of blame upon his mother.

Madam Washington, having gained her heart's desire, could not now do too much for George. He was already far advanced beyond Mr. Hobby's school, and his mother determined to have a tutor for him. Nothing was too good for him now; his tutor must be a university man, with every qualification in family and manners as well as learning. But there was no such person within reach, and communication in those days being slow and uncertain, there seemed no immediate chance of finding one. George went his way calmly, but with his disappointment eating into his heart. He studied surveying, in which he was already proficient, with Mr. Hobby, but he did nothing else. Even his beloved hunting and shooting palled upon him. He would spend the day at work, having Mr. Hobby's help in the afternoon, and at night he would work out at home what he had done during the day. Mother and son never failed in courtesy and even affection for each other; indeed, Madam Washington lavished affection upon him in a manner hitherto unknown to him; but there was a little shadow between them.

Heretofore George had not escaped being lectured for his youthful shortcomings, but no fault was ever found with him now. Even Billy's laziness was excused, and he might be as idle as he pleased; like his young master, he enjoyed a complete immunity from fault-finding. This was not a natural or a healthy way for the mother and son to live; and one day, when George walked in and laid a letter from Lord Fairfax in his mother's hand, saying, simply, "I think I should like that, mother," Madam Washington, with one sharp pang, felt that they must part—at least for a while.

The letter was brief, and had no mention of the warrant in the navy, by which George subtly understood that Lord Fairfax knew it was a delicate subject, and would say nothing about it. The Earl wrote, however, that he had determined to have his lands across the mountains surveyed

during the coming summer, and offered George for it a sum of money so large that to the boy's unsophisticated mind it seemed a fortune. But Lord Fairfax stipulated that George should have a license from the State of Virginia, as his surveys would no doubt often be called in question, and there must be a recorded proof of his efficiency.

Madam Washington sighed deeply, yet there was no doubt that he must go. He would be sixteen within a few days, and he was already as developed in mind and body as a young man of nineteen. Her plans for his further education seemed impossible to realize, and it was plain there was but one thing to do—to let him go. She told him so that night, and the first gleam of sunshine came into his face that she had seen since the day after his return home. Betty's comment was like her.

"If you want to go, George, I want you to go; but it will be doleful at Ferry Farm without you."

George immediately made preparations for his examination in surveying, and having passed it successfully and got his certificate, he was ready to start on his journey as soon as the spring should open. He wrote to his brother Laurence, stating his plan, and saying he would spend a night at Mount Vernon on his way. Laurence had shown the same consideration for George's feelings that Lord Fairfax had, and, in reply to the letter returning the midshipman's warrant, had merely said that he regretted he had not known of Madam Washington's determination sooner. One sentence at the end touched George: "Your little niece is well, but she is but a frail child, and I have a presentiment that Mount Vernon will never come to any child of mine. For that reason, as you will some day be master of this place, I would like to have you here as often and as long as your mother can spare you. My own constitution is delicate, and nothing is more probable than that you will have Mount Vernon for your own before you are of age."

Madam Washington made the preparations for George's departure with a steady cheerfulness that belied her sad heart. She herself proposed that he should take Billy along. She offered him such a considerable sum of money that George knew she must be depriving herself of many things, and refused to take it all. In every way there was a strong though silent purpose to make up to him for her one moment of weakness. George felt this, and when, on the morning of his departure, his mother bade him good-by, with a smile on her pale lips, he felt a softening of the heart towards her that lasted not only during this separation, but through all the coming years, with their tremendous events.

Little Betty wept torrents of tears, protesting all the time. "Dear George, I am glad for you to go—I don't want you to stay—I can't help crying a little, though."

George held her in his arms with a full heart, and wished that he had words to tell her how much she was to him; but Betty understood well enough. When the last farewells were said, and George was out of sight of his mother's brave smile and Betty's tears, a sudden revulsion of feeling came to him, as it does to all healthy young natures. He had got to the very extremity of his despair, and there was a strong reaction. He was essentially a boy of action, and action was now before him. Indeed, he was no longer a boy, but a man, with responsibilities upon him that seldom fall to young people of his years. He had his surveyor's license in his pocket, and upon the use he made of it might depend not only issues of property, but of peace and war; because he knew that the unsettled state of the frontier was the real reason why Lord Fairfax meant to have the wild lands in his grant surveyed. The day was bright, it was in the spring-time, and he was well mounted on a good horse. Billy, riding a stout cart-horse and carrying the saddle-bags, was behind him, and Rattler was trotting by his side. Things might be worse, thought George, as he struck into a canter, and wondered that his heart was so blithe. He would see his brother and sister that night, and little Mildred, and in a few days more he would be again at Greenway with the Earl and Lance; and he

would have all the books he wanted to read, and fencing whenever he liked. He wondered how much he had forgotten of it; he had not fenced since leaving Mount Vernon at Christmas. But neither had he read or done anything else, it seemed to George, so blank was the time from the day he came home until then. Billy hankered after the flesh-pots of Mount Vernon, where things were conducted on a much grander scale than at the simple Ferry Farm homestead. George heard him chuckling to himself, and turning in the saddle, asked,

"What pleases you so, Billy?"

"Tuckey, suh," answered Billy, promptly, "wid sassages roun' dee necks—an' oshters an' sp'yar-ribs an' chines an' goose, an' all dem t'ings dee black folks gits in de kitchen at Mount Vernon."

It was a good forty-five miles to Mount Vernon, but George made it by eight o'clock that night.

His brother and sister were delighted to see him, and little Mildred had not forgotten him. After a traveller's supper, George told them all his plans. He passed quickly over the giving up of his midshipman's warrant, merely saying, "My mother begged me not to leave her for the sea, and I consented. But," he added, after a pause, "it nearly broke my heart."

He was distressed to see his brother looking so pale and thin, and still more so at the despondent tone Laurence took about himself. He would have had George go into the study, and there with him discuss the present state of the place and its future management, as if he were certain that one day it would be George's; but this the boy refused.

"No, brother," he said, "I can only inherit Mount Vernon through misfortune to you and yours; and do you suppose I like to think about that? Indeed I do not; and I neither think nor care about what you do on the place, except that it shall be for your own satisfaction."

The next morning George was off, much to the regret of his brother and sister, and also of Billy, who had promised himself a regular carnival in the Mount Vernon kitchen.

The road was the same that George had taken nearly five months before, on his first expedition to Greenway Court. Then it had been at the fall of the leaf, and now it was at the bursting of the spring. Already the live-oaks and poplars were showing a faint and silvery green, and in sheltered sunny spots grass was sprouting. The water-courses were high from the melting of the snow, and fording them was not always without difficulty, or even danger. At every mile that George travelled his mind and heart gained a better balance by quick degrees. He was sorry to be parted from his mother and Betty, but he was at a time of life when he must try his own strength, and he was the better for it. He stopped at the same taverns that he had halted at when with Lord Fairfax. Billy proved himself to be an excellent hostler as well as valet, and George did not mean to forget mentioning to his mother, when he should have an opportunity of sending a letter, how extremely useful Billy was. On the fourth day, being well up in the mountains, they came to Lord Fairfax's coach-house, as it was called; but instead of stopping, George pushed on to Greenway Court, much to Billy's disgust, who had no taste for long journeys on traveller's fare. On a March night, that, although cool, had a touch of spring in the air, and under a glorious moon, George rode up to the door at Greenway Court, and joyfully dismounted. Lord Fairfax did not know the exact day to expect him, but knew he would arrive about that time. When George's loud rat-tat resounded upon the great oak doors, it seemed the most natural thing in the world to have them opened by old Lance, who said, as if he had seen George half an hour before:

"Good-evening, Mr. Washington; my lord is expecting you. Billy, take the horses around to the stable."

George walked in, and almost ran into the Earl's arms. Lord Fairfax was overjoyed to see him, and although he did not say much, his pleasure shone in his eyes. George's room was ready for him; there was a fine young half-thoroughbred in the stables that was waiting for George's saddle and bridle to be put on him; Lance had some bears' paws for his supper whenever he should arrive; there were some books on surveying imported from England for him. Had he been Lord Fairfax's son and heir he could not have been received with greater consideration. The Earl could not do enough for him. It was:

"Lance, is Mr. Washington's room prepared for him?"

"Yes, sir. It has been ready for a week."

"And, Lance, Mr. Washington will probably want you in the morning in the armory."

As soon as supper was over, George displayed proudly his license as surveyor, and would have plunged into the affair of the surveys at once, but Lord Fairfax gave the first intimation then that he did not consider George a full-fledged man.

"Never mind for to-night, George. Very young gentlemen like you are apt to go at things like a hunter at a five-barred gate, but you can wait awhile. Besides, you must go to bed early after your journey, so as to get sleep—a thing that growing boys cannot do without."

George felt several years younger at this speech, and blushed a little for his mannish airs, but the Earl's advice about going to bed was sound, and in five minutes after finding himself in the great high-post bed he was sleeping the sleep of healthy and active boyhood.

[TO BE CONTINUED.]

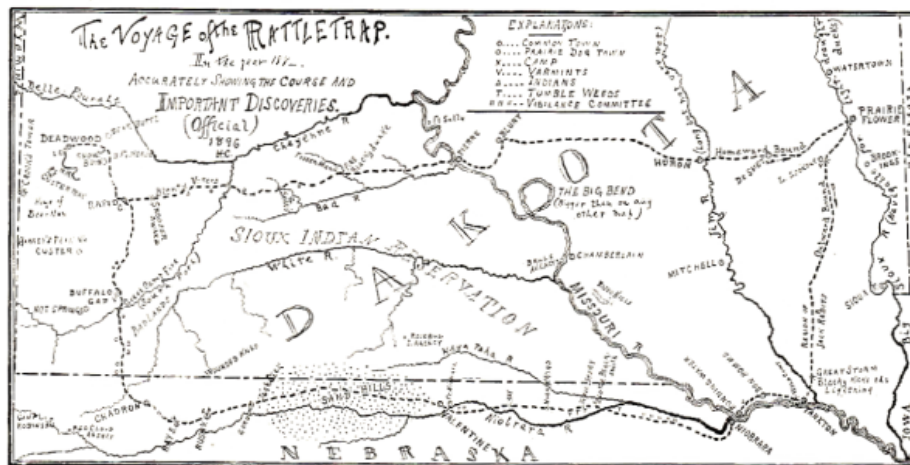
THE JUJUBE MAN.

BY ALBERT LEE.

At the foot of Spice Lane, in Carameltown,
Lives a funny little jujube fellow;
His body is red and his legs are brown,
And his hands and his feet are yellow.

This jujube man has a nice little farm,
With a rooster, a hen, and a hog,
All well watched over and kept from harm
By a little red jujube dog.

But the jujube man does not like the heat,
For it almost makes him melt,—
And his head bends over 'most down to his feet,
And his toes bend up to his belt.



THE VOYAGE OF THE "RATTLETRAP."

BY HAYDEN CARRUTH.

IV.

We were a good deal disappointed in not getting over into Nebraska, because we had seen enough of Dakota, but there was no help for it. A log had got caught in the paddle-wheel of the ferry-boat and wrecked it, and there was no other way of crossing.

"Old Blacky could swim across," said Jack, "but Browny would go to sleep and drown."

It is rather doubtful, however, about even Blacky's ability to have swum the river, since it was a half-mile wide, and with a rather swift current. In the afternoon we walked back to Yankton and bought the biggest felt hats we could find, with wide and heavy leather bands. We knew that we would now soon be out in the stock-growing country, and that, as Jack said, "the cowboys wouldn't have any respect for us unless we were top heavy with hat."

We were camped on the high bank of the river, opposite a farm-house. It was getting dusk when we got back to the wagon, with our heads aching from our new hats, which seemed to weigh several pounds apiece. Jack, as cook, announced that there was no milk on hand, and sent Ollie over to the neighboring house to see if he could get some. Ollie returned, and reported that the man was away from home, but that the woman said we could have some if we were willing to go out to the barn-yard and milk one of the cows. The others decided that it was my duty to milk, but I asked so many foolish questions about the operation, that Jack became convinced that I didn't know how, and said he would do it himself. We all went over to the house, borrowed a tin pail from the woman, and went out to the yard.

We found about a dozen cows inside, of various sizes, but all long-legged and long-horned.

"Must be this man belongs to the National Trotting-Cow Association," said Jack, as he crawled under the barbed-wire fence into the yard. "That red beast over there in the corner ought to be able to trot a mile in less than three minutes."

He cautiously went up to a spotted cow which seemed to be rather tamer than the rest, holding out one hand, and saying, "So, bossy," in oily tones, as if he thought she was the finest cow he had ever seen. When he was almost to her she looked at him quickly, kicked her nearest hind foot at him savagely, and walked off, switching her tail, and shaking her head so that Ollie was afraid it would come off and be lost.



TOP HEAVY WITH HATS.

"Can't fool *that* cow, can I?" said Jack, as he turned to another. But he had no better luck this time, and after trying three or four more he paused and said:

"These must be the same kind of cows Horace Greeley found down in Texas before the war. When he came back he said the way they milked down there was to throw a cow on her back, have a nigger hold each leg, and extract the milk with a clothes-pin."

But at last he found a brindled animal in the corner which allowed him to sit down and begin. He was getting on well when, without the least warning, the cow kicked, and sent the pail spinning across the yard, while Jack went over backwards, and his new hat fell off. There was one calf in the yard which had been complaining ever since we came, because it had not yet had its supper. The pail stopped rolling right side up, and this calf ran over and put his head in it, thinking that his food had come at last. Jack picked himself up and ran to rescue the pail. The calf raised his head suddenly, the pail caught on one of his little horns, and he started off around the yard, unable to see, and jumping wildly over imaginary objects. Jack followed. A cow, which was perhaps the mother of the calf, started after Jack. The family dog, hearing the commotion, came running down from the house and began to pursue the cow. This wild procession went around the yard several times, till at last the pail came off the calf's head, and Jack secured it. Then he picked up his hat, the brim of which another calf had been chewing, rinsed out the pail at the pump, and tried another cow.

This time he selected the worst-looking one of the lot, but to the surprise of all of us she stood perfectly still, only switching him a few times with her tail. As soon as he got a couple of quarts of milk he stopped and came out of the yard. Ollie and I had, of course, been laughing at him a good deal, but Jack paid no attention to it. As we walked toward the house he said:

"Well, there's one consolation; after all of that work and trouble the woman can't put on the face to charge us for the milk." A moment later he said to her, "I've got about two quarts; how much is it?"

"Ten cents," answered the woman. "Didn't them cows seem to take kindly to you?"

"Well, they didn't exactly crowd around me and moo with delight," replied Jack, as he handed over a dime with rather bad grace.

That evening a neighbor called on us as we sat about our camp fire, and we told him the experience with the cows.

"Puts me in mind of the time a fellow had over at the Santee Agency a year or so ago," said our visitor. "There's a man there named Hawkins that's got a tame buffalo cow. Of course you might as well try to milk an earthquake as a buffalo. Well, one day a man came along looking for work, and Hawkins hired him. Milking-time came, and Hawkins sent the man out to milk, but forgot to tell him about the buffalo. The man was a little green, and it was sort of dark in the barn, and the first thing he tried to milk was the buffalo cow. She kicked the pail through the window, smashed the stall, and half broke the man's leg the first three kicks. He hobbled to the house, and says to Hawkins, 'Old man, that there high-shouldered heifer of yours out there has busted the barn and half killed me, and I reckon I'll quit and go back East, where the cows don't wear sleigh-robies and kick with four feet at once.'"

Bright and early the next morning we got off again. Nothing of importance happened that day. We were travelling through a comparatively old-settled part of the country, and the houses were numerous. A young Indian rode with us a few miles, but he was a very civilized sort of red man. He had been at work on a farm down near Yankton, and was on his way to the Ponca Reservation to visit his mother. As an Indian he rather disgusted Ollie.



THE COW THAT WORE A SLEIGH-ROBE, AND KICKED WITH ALL FOUR FEET.

"If I was a big six-foot Indian," he said, after our passenger had gone, "I think I'd carry a tomahawk, and wear a feather or two at least. I don't see what's the advantage of being an Indian if you're going to act just like a white man."

We camped that night in a beautiful nook in a bluff near a little stream. The next day we reached Running Water. The ferry-boat was a little thing, with a small paddle-wheel on each side operated by two horses on tread-mills. A man stood at the stern with a long oar to steer it. The river was not so wide here as at Yankton, but the current was swifter, which no doubt gave the place its name. It looked very doubtful if we would ever get across in the queer craft, but after a long time we succeeded in doing so. It gave us a good opportunity to study the water of the river, which looked more like milk than water, owing to the fine clay dissolved in it. The ferryman thought very highly of the water, and told

us proudly that a glass of it would never settle and become clear.

"It's the finest drinking-water in the world," he said. "I never drink anything else. Take a bucket of it up home every evening to drink overnight. You don't get any of this clear well-water down me."

We tasted of it, but couldn't see that it was much different from other water.

"Boil it down a little, and give it a lower crust, and I should think it would make a very good custard pie," said Jack.

We found Niobrara to be a little place of a few hundred houses. We went into camp on the edge of the town, where we staid the next day, as it was Sunday. Early Monday morning we were out on the road which led along the banks of the Niobrara River. We were somewhat surprised at the smallness of this stream. It was of considerable width but very shallow, and in many places bubbled along over the rocks like a wide brook. We spoke of its size to a man whom we met. Said he:

"Yes, it ain't no great shakes down here around its mouth, but you just wait till you get up in the neighborhood of its head-waters. It's a right smart bit of a river up there."

"But I thought a river was usually bigger at its mouth than at its source," I said.

"Depends on the country it runs through," answered the man. "Some rivers in these parts peter out entirely, and don't have no mouth a' tall—just go into the ground and leave a wet spot. This here Niobrara comes through a dry country, and what the sun don't dry up and the wind blow away the sand swallows mostly, though some water does sneak through, after all; and in the spring it's about ten times as big as it is now. The Niobrara goes through the sand hills. Anything that goes through the sand hills comes out small. You fellers are going through the sand hills—you'll come out smaller than you be now."

This was the first time we had heard of the sand hills, but after this everybody was talking about them and warning us against them.

"Why," said one man, "you know that there Sarah Desert over in Africa somewhere? Well, sir, that there Sarah is a reg'lar flower-garden, with fountains a-squirting and the band playing 'Hail Columbia,' 'longside o' the Newbraska sand hills. You'll go through 'em for a hundred miles, and you'll wish you'd never been born!"

This was not encouraging, but as they were still several days' travel ahead, we resolved not to worry about them.

About the middle of the afternoon we came upon a great level prairie stretching away to the west as far as we could see. There seemed to be but few houses, and the scattering fields of corn were stunted and dried up. It had apparently been an extremely dry season, though the prospects for rain that night were good, and grew better. It was hot, and a strong south wind was blowing. Night soon began to come on, but we could find no good camping-place. We had not passed a house for four or five miles, nor a place where we could get water for the horses. As it grew dark, however, it began to rain. It kept up, and increased to such an extent that in half an hour there were pools of water standing along the road in many places, and we decided to stop. It was wet work taking care of the horses, but the most discouraging thing was the report from the cook that there was no milk with which to make griddle-cakes for supper, and as he did not know how to make anything else, the prospect was rather gloomy. But through the rain we finally discovered a light a quarter of a mile away, and Ollie and I started out to find it. Jack refused to go, on the plea that he was still lame from his Yankton trip after milk.

We blundered away through the rain and darkness, and after stumbling in a dozen holes, running into a fence, and getting tangled up in an abandoned picket-rope, at last came up to the house. It was a little one-room board house such as the settlers call a "shack." The door was open, and inside we could see a man and woman and half a dozen children and a full dozen dogs. We walked up, and when the man saw us he called "Come in!" tossed two children on the bed in the corner, picked up their chairs, which were home-made, and brought them to us.

"Wet, ain't it?" he exclaimed. "Rainy as the day Noah yanked the gang-plank into the Ark. I was a-telling Martha there was a right smart chance of a shower this afternoon. What might you-uns' names be, and where might you be from, and where might you be going?"

We told him all about ourselves, and he went on:

"Rainy night. Too late to help the co'n, though. Co'n's poor this year; reckon we'll have to live on taters and hope. Tater crop ain't no great shakes, though. Nothing much left but hope, and dry for that. Reckon I'll go back to old Missouri in the spring, and work in a saw-mill. No saw-mills here, 'cause there ain't nothing to saw. Hay don't need sawing. Martha," he added, turning to his wife, "was it you said our roof didn't need mending?"

"I said it did need it a powerful sight," answered the woman, as she put another stick of hay in the stove, and a stream of rain-water sputtered in the fire.

"Mebby you're right," said the man. "There's enough dry spots for the dogs and children, but when we have vis'tors somebody has got to get wet. Reckon I oughter put on two shingles for vis'tors to



"WET, AIN'T IT?"

set under. You fellers will stay to supper, of course. We 'ain't got much but bacon and taters, but you're powerful welcome."

"No," I said, "we really mustn't stop. What we wanted was to see if we couldn't get a little milk from you."

"Well, I'll be snaked!" exclaimed the man. "That makes me think I 'ain't milked the old cow yet."

"I milked her more'n two hours ago, while you was cleaning your rifle," said his wife.

"That so?" replied the man. "Where's the milk?"

The woman looked around a little. "Reckon the dogs or the young 'uns must 'a' swallowed it. 'Tain't in sight, nohow."

"Oh, we can milk 'er again," exclaimed the man. "Old Spot sometimes comes down heavier on the second or third milking than she does on the first."

He took a gourd from a shelf, and told us to "come on," and started out. He wore a big felt hat, but no coat, and he was barefooted. Just outside the door stood a bedstead and two or three chairs. "We move 'em out in the day-time to make more room," explained the man. The rain was still pouring down. The man took our lantern and began looking for the cow. He soon found her, and while I held the lantern, and Ollie our jug, he went down on his knees beside the cow and began to milk with one hand, holding the gourd in the other. The cow stood perfectly still, as if it was no new thing to be milked the second time. We had on rubber coats, but the man was without protection, and as he sat very near the cow a considerable stream ran off of her hip bone and down the back of his neck. When the gourd was full he poured it in our jug, and at my offering to pay for it he was almost insulted. "Not a cent, not a cent," he exclaimed. "Al'ays glad to 'commodate a neighbor. Good-night; coming down in the morning to swap hosses with you."

He went back to the house, and we started for the wagon.

"He wouldn't have got quite so wet if he hadn't kept so close to the cow," said Ollie, as we walked along.

"What he needs," said I, "are eave-troughs on his cow."

[TO BE CONTINUED.]

HOW CAPTAIN JACK GOT HIS TITLE.

"Captain Jack," said Tommie, as he and Bobbie drew near to the Old Sailor at the sea-shore, shortly after their arrival, "you've told us a great many stories, but you never told us how you came to be a Captain. Was it for bravery in battle?"

"No, my lad," replied the old Captain. "I've been brave enough in battle to be an Admiral, but I never got no promotion for it. It was indoorance won my title for me."

"Endurance?" said Bobbie. "That's as good as bravery, isn't it?"

"Better," said Captain Jack. "A great deal better. A great many brave people give out when they oughtn't to, but indoorin' people never gives out."

"Nor in, neither," said Tommie, "I guess."

"I guess likewise," said Captain Jack. "It wuz this way: In eighteen seventy-one—no, I guess it was eighteen seventy-three—no—waal I never—when was it?"

"Make it 1874," said Bobbie. "Three and one make four."

"That's when it was," said Captain Jack. "In eighteen seventy-four I shipped as a able-bodied seaman before the mast with Captain William Bilkes, of East Gloucester, Massachusetts, of the brig *Peter J.*, of Nantucket. The *Peter J.* was a pretty good boat. They called her a brig, but she wasn't nothin' in particular, as far as I could see. She was a composite boat—like them fortygrafts. The owners of her bought her stern in New York, an' fastened it onto the bow of a wrack they'd purchased in a junk-shop at Plymouth. The rudder wuz a relict of a defunct Spanish man-o'-war, an' the masts wuz bought at a bargain sale o' ship stuffs at Phillydelphy. Whar the cabin come from I dun'no', but it was amatoor from way back.

"When I fust seed the ship I says, No, I don't want none o' her in mine. I'm fond o' swimmin', but I wants it as a diwersion an' not fer bizness. But Cap'n Bilkes he says to me, says he: 'Jack you're the best sailor afloat, an' I needs yer. Come with me, an' I'll give yer two thousand dollars a month!'

"'Cap'n,' says I, 'that ain't what I gen'rally gets, but to oblige ye I'll come at them figgers;' an' I went, not askin' at all where we was a-goin' to go to.

"'Waal, we sets out, me before the mast with the others, an' the Cap'n an' two mates, four midshipmen, three soupycargoes, an' others behind the mast.

"First day out, down comes the Cap'n with the mumps. Dies. Chucked overboard.

"Second day out, down comes the two mates with measles. Dies. Buried at sea.

"Third day out, down comes the soupycargoes with whoopin'-cough. Dies. All's over.

"Fourth day out, down comes the hull crew, except me, with shycumotis, due to havin' eat too much tomatoes. Dies. Nothin' left aboard but me with the *Mary Jones*—"

"*Peter J.*," said Bobbie.

"The *Peter J.*," observed Captain Jack. "Git 'em mixed sometimes; they's so many boats, I 'ain't more trouble to— There was me all alone by myself aboard the *Henry Q.* to bring her into port, loaded as she was with olives an' fried potatoes.

"It was a tarrable responsibility, but I took it on. So, my boys, there bein' no soupycargoes, nor no mates, nor no cap'ns, nor no nothin' save me an' the decks onto the ship, I 'p'inted myself commander-in-chief, an' thar ye be."

"It's very interesting," said Tommie.

"You bet it is!" said Captain Jack; "but it ain't half so interesting to me as a box o' imported cigars would be."

And the boys walked off, and later on Captain Jack received a box of imported cigars—"just," as Tommie said, "to interest him."

A SMALL ELECTRIC-LIGHT OUTFIT.

HOW TO MAKE THE BATTERY AND HOW TO SET IT UP.

BY THOMAS R. TALTAVALL.

All young readers of HARPER'S ROUND TABLE are probably familiar with the appearance of the electric light—that wonderful little glass bulb that we see in stores, hotels, theatres, etc.—and no doubt many have wondered what causes the hairlike loop inside the bulb to become so brilliantly luminous. Electricity does it; but no one knows what electricity is, not even the most advanced scientific thinker of the day. We know how to produce it and how to handle it, but further than that we are still in ignorance.

The electricity necessary to light an incandescent lamp, such as we see in stores, etc., is generated in a machine called a *dynamo*. The dynamo is driven by a steam-engine, and produces electric current in large quantities for electric-lighting on a large scale. Electricity can also be produced by a battery, but in very small quantity as compared with that produced by a dynamo; and in order to light a house by electricity from a battery, so many cells would be required that it would be decidedly inconvenient and troublesome to keep them in order.

It may interest the boys among the readers of HARPER'S ROUND TABLE to know that with a little ingenuity a battery can be constructed by themselves which will give sufficient current to light a small electric lamp. Such an outfit is full of interest and instructive to young people, and its use creates a desire for a wider knowledge of this most fascinating subject.

Electric lamps are made of all sizes, from the size of a pea to that of the lamp we see every day and are most familiar with, and they are used for a great variety of purposes. The lamps referred to here are those having a pear-shaped glass bulb, and which are known as incandescent lamps. The kind we see in the streets, giving a very powerful light, are called arc-lights. With this class of lights, however, we will have nothing to do here. The object of this article is to give plain directions for the construction of a battery to light a small electric lamp. Some boys may think that because electricity is something mysterious it must be very difficult to produce; but that is not the case. A battery is easily made, and by following the directions here given we can produce just the same kind of light we see in stores and other public places, only on a smaller scale.

We must provide three things, namely, the battery, the lamp, and the wire to conduct the current from the battery to the lamp.

First, the battery. A cell of battery, such as we shall need, is made up of four constituent parts—the glass jar, a rod of zinc, a couple of carbon rods, and the solution in which the carbons and zincs are to be immersed. Two such cells will be needed to give a brilliant light.

For our purpose ordinary glass tumblers will answer very well for the jars. Tumblers are suggested, because almost every boy can obtain his mother's consent to use a couple for this purpose.

The next things to provide are the zincs and carbons. The zinc may be of any shape, flat, square, or round, but we have selected the round form for our battery because it is easily obtained, and more easily handled and prepared by the amateur for an experimental battery. All electrical-supply

houses keep Leclanche zincs, which are rods of that metal about $\frac{3}{8}$ of an inch in diameter and 9 inches long. One such zinc cut in two will give us two pieces each $4\frac{1}{2}$ inches long, which will be ample for our battery.

Carbons, like zincs, are made in many forms, but for our purpose we have also selected the round shape because they are likewise more easily obtained. Such carbons as are used in electric-arc street lights will answer very well indeed. These carbons are usually plated with a thin coating of copper, which must be removed before we can use them for our battery. This can often be done by scraping the copper off. Should that fail to completely remove it, nitric acid will; but I would not advise using acid if the copper can be removed the other way. Nitric acid is dangerous stuff to have around on account of the fumes it gives off, and its corrosive propensities when it comes in contact with fabrics and almost every other substance.

Having procured your carbons, cut them off the same length as the zincs, which can be easily done with a scroll or any other small saw. Be careful, though, because carbon is rather brittle, and will break rather than bend. The zincs and carbons should be of the same length, and should be at least an inch higher than the top of the tumbler. Two carbons and one zinc will be required for each cell.

The next step in our work is to provide means for suspending the zincs and carbons in the tumblers. This will require a little carpenter-work. Get two pieces of well-seasoned and very dry wood (any kind will do), each about $1\frac{1}{2}$ inches wide, 4 inches long, and $\frac{3}{8}$ or $\frac{1}{2}$ an inch thick, and bore three holes in each in the manner shown in Fig. 3. The holes should be, as nearly as possible, of the same diameter as the zincs and carbons, and the middle one should be midway between the ends of the wooden "hanger." The zincs and carbons are to be placed in these holes, and suspended in the tumblers as shown in Fig. 4, the zinc rod being placed in the centre, with one carbon rod on each side.

We are now ready to attach the wires to the zincs and carbons for the purpose of conveying the current from the battery to the lamp. Use No. 18 braided wire, which can be obtained of any dealer in electrical supplies. Any one caring to go to the extra expense, however, can get a silk-covered cord which is very neat and convenient, because two wires are twisted together, forming one cord. Each conductor of this cord is made up of several fine copper wires, instead of one solid wire, which makes it very flexible. But whether the cord or separate single wires are used, the connections with the zincs and carbons are the same. It is understood, of course, that it is necessary to have two wires to connect the battery and lamp. One end of each of the wires is connected with the lamp, and the other ends with the battery—one with the zincs and the other with the carbons.

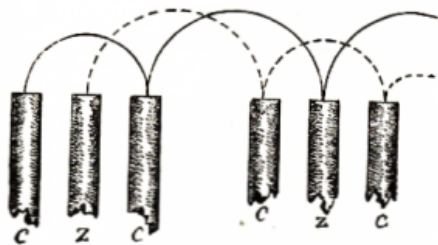


FIG. 1.

Fig. 1 shows how these battery connections must be made. One of the lamp wires is connected with the two carbons of cell No. 1 and the zinc of cell No. 2, and the other wire is connected with the zinc of cell No. 1 and the two carbons of cell No. 2.

To make these connections, have a sufficient length at one end of each of the wires to wrap four or five times around the carbons and zinc, allowing a little slack between them. Wrap as firmly as possible so as to insure a tight joint, as nothing is more wasteful of current than a loose

connection. We need all the current for our lamp. Before wrapping the wire it should be well scraped and cleaned, in order to secure a good connection with the zinc and carbons. A dirty connection is as bad as a loose one.

Having connected the two wires with the carbons and zincs in the manner shown, in Fig. 1, we are now ready to connect the lamp. Remove the braid, or covering, of the free ends of the two wires for about two or three inches, and after scraping and cleaning the ends of the wire, twist them tightly with the two wires projecting from the lamp; then all the connections are made. If you now place the zincs and carbons in the solution (which I will presently refer to more particularly), your battery will produce a current which will make the little lamp give a light as sparkling as a diamond. The lamp is the most important part of the outfit, and cannot be made by amateurs. It must be bought of a dealer.

Fig. 2 is an illustration of a lamp that will give a light of $\frac{1}{2}$ candle-power or more with the battery described above. This illustration is the actual size of a $\frac{1}{2}$ candle-power lamp. The lamp is a small glass bulb, inside of which is a short length of carbon in the shape of a small arch. This carbon is connected with the wires running through and outside of the glass bulb, and when the current flows through the wires the carbon becomes "incandescent"—that is, white-hot. This carbon is in a vacuum, the air having been exhausted from the bulb in the process of manufacture. The vacuum is essential. Should there be none, and the space be filled with air, the carbon would be destroyed by the oxygen in the air the moment the current rendered it incandescent.



FIG. 2.

In handling the lamp while making connections with the battery wires, care must be taken that the lamp wires are not broken off close to the glass by too much twisting. Should this happen, the lamp would be rendered useless, because then we could not get a connection with the carbon filament.

All the connections having been made, everything is ready to charge the battery. Charging means to fill the cells with the proper chemical solutions which, in their action upon the zinc and carbons, produce the electric current. The chemicals necessary to make the solution for the battery can be bought at any drug-store, but those of our young friends who live in cities can buy the battery fluid all ready prepared at any electrical-supply house, if they do not care to bother with making it themselves. I would advise making it fresh, however, because it gives young experimenters some experience and something to think about. The fluid is called in the trade "electropoin fluid," and is sold by the pint, quart, gallon, or any other quantity, by the regular dealers. All batteries using this fluid are generally called "Electropoin" batteries.

To make this fluid the following ingredients are necessary: bichromate of potash, sulphuric acid, and water. Bichromate of potash comes in lumps, and is of a dark red or wine color; it can be bought at any drug-store. Sulphuric acid can also be bought at any drug-store; it must be handled very carefully, as it has a disagreeable habit of burning, and otherwise destroying almost everything it comes in contact with. Glass is one of the few substances it does not attack, therefore it is safe in a glass bottle.



FIG. 3.

The following is a receipt for preparing electropoin fluid: take $1\frac{1}{2}$ lbs. of bichromate of potash, and after crushing the lumps as fine as possible, dissolve it in 2 quarts of boiling water. After this solution has become cold, add to it a solution composed of 1 quart of sulphuric acid and 3 quarts of water, and thoroughly mix the two. After the mixture is cool it is ready for use. This will be enough for several charges of the battery. A smaller quantity may be

made by proportionate reduction in quantity of the several ingredients.

It is a very strange fact that pouring sulphuric acid into water produces no different effect than if the acid were so much water, but if the water is poured upon the acid a greatly different effect takes place. Heat is very rapidly developed, causing the liquids to boil violently, and sputtering and scattering in every direction. Care must be taken, therefore, to *pour the acid into the water*.

When the electropoin fluid is cool we can go ahead and set up our battery. Place the zincs and carbons in position in their wooden supports, Fig. 3, and suspend them in the tumblers, as shown in Fig. 4; pour the bichromate solution into the tumblers until it reaches within $\frac{3}{4}$ inch or 1 inch of the top. The wire connections should be made with the zinc and carbons before the battery is set up, because to attempt to make them after the solution is in the tumblers you cannot help spilling the fluid. And I might caution my young friends right here to be careful not to spill any of the solution on their clothes or on the carpet, as it eats holes in fabrics.

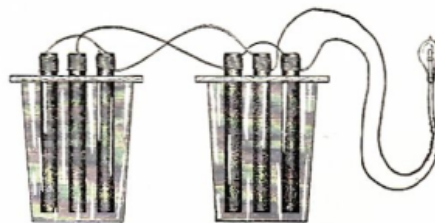


FIG. 4.

Now for the final touches. If we now connect the lamp to the battery wires it will give a brilliant light the moment we make the last connection, and continue to burn until the current is broken, which may be done by lifting the elements (as the zincs and carbons are called) out of the fluid. After a while, if the current is kept on for some time, the light will begin to diminish in brilliancy. This is due to exhaustion of the liquid in the tumblers, and when the light gets dim the solution should be renewed.

In connecting your lamp with the battery wires simply twist the lamp and battery wires together firmly but gently. All of the connections are shown very clearly in Fig. 2.

When the light is not desired it is best to remove the elements from the tumblers, so as to save the zinc. The acid in the solution dissolves the zinc.

An outfit of this sort costs about as follows: one Leclanche zinc (sawed in two), 8 cents; two plain electric-light carbons, 20 cents; one pound sulphuric acid (about one-half pint), 15 cents; one pound bichromate potash, 25 cents; twenty-five feet No. 18 wire, 15 cents; one-half candle-power lamp, \$1; total cost, \$1.83.

I have not included in this estimate the cost of the two tumblers or the wooden holders, because tumblers can always be had around the house, and no boy needs to be told where he can find wood. But to be on the safe side we will assume that tumblers and all have to be bought; then \$2 will cover the cost easily.

Such an outfit as this is very interesting and instructive to every boy—and girl, too, for that matter—and what can be more fascinating than to be able to produce a beautiful electric light so easily?



The sixth National Interscholastic Tennis Tournament was played last week at Newport, and resulted in a victory for Reginald Fincke, of the Hotchkiss School, winner of the Yale Interscholastic Tournament. His victory was more or less of a foregone conclusion on the form he had displayed in his early spring work; and the opponents he had to meet were none of them very formidable, the strongest, Beggs of Lawrenceville, not being present. This gave Walton of Colombia, the weakest man of the lot, a chance to get into the semi-finals, where he succumbed to the Harvard interscholastic representative in three straight and uninteresting sets.

INTERSCHOLASTIC TENNIS CHAMPIONSHIP SERIES.

Year.	Played at.	Winner.	School.
1891.	Cambridge.	R. D. Wrenn.	Cambridge Latin.
1892.	Cambridge.	M. G. Chace.	Univ. Grammar, Prov.
1893.	Newport.	C. R. Budlong.	High, Providence.
1894.	Newport.	W. G. Parker.	Tutor, New York,
1895.	Newport.	L. E. Ware.	Roxbury Latin.
1896.	Newport.	R. Fincke.	Hotchkiss.

Fincke drew Turner of Chicago in the preliminaries, and defeated him, 6-4, 6-2, 6-8, 6-3. He then defeated Willing, the U. of P. interscholastic champion, 6-2, 6-1, 5-7, 6-4. His hardest match was with Edwards of English High-School. Edwards made a good brace in the third set, taking it 6-2, but he was unable to maintain this form, and although he did good work in the last set, he was unable to end better than 4-6. This gave the championship for 1896 to Fincke, 6-2, 6-4, 2-6, 6-4. A full summary of the play will be found on the next page; and the championship list now stands as shown in the table given above.

SIXTH ANNUAL NATIONAL INTERSCHOLASTIC TENNIS TOURNAMENT, NEWPORT, AUGUST 20, 21, 22, 1896.

Preliminary Round.	Semi-Final Round.	Final Round.	Interscholastic Champion.
	J. K. Willing (U. of Pa.),	}	
R. Fincke (Yale),	} Fincke, 6-4, 6-2, 6-8,	} Fincke, 6-2, 6-1, 5-7, 6-4.	}
L. H. Turner (Univ. of Chicago).	}		}
			} Fincke, 6-2, 6-4, 2-6, 6-4.
C. W. Beggs (Princeton),	}		}

J. McL. Walton
(Columbia).

} Walton, by default. } Edwards, 6-1, 6-0, }
6-3.

Y. M. Edwards
(Harvard). }

It is sincerely to be hoped that the unfortunate differences between those stanch old rivals and worthy opponents, Exeter and Andover, will be brought to a friendly close this year. Since 1893 these two schools, who used to meet annually in football, baseball, and tennis, have not had any athletic intercourse whatever with one another, and all because, in their great rivalry and desire to beat one another, one of them certainly (and possibly both) overstepped the bounds of athletic ethics, and was guilty of practices which, by a little consideration and forethought, might have been avoided.

It is improbable that any good can be accomplished by going over the causes which brought about this rupture between the two schools, for stirring up mud and opening old wounds are of small benefit. Those who know what the causes were can do nothing better than to forget, and those who do not know what the trouble was will certainly find that here is a case where ignorance is bliss.

For the past year or more I have been in correspondence more or less actively with Exeter and Andover men, graduates and students and members of the two Faculties, and the general opinion among all seems to be that it would be a good thing to have the games renewed; and while both sides seem willing to meet again, neither side seems anxious to make the first advances. Both schools have, or think they have, grievances; and each thinks the other should lay aside its pride or stubbornness and make the friendly advances.

This situation might remain for an indefinite period if some strong-minded individual or individuals did not step in and say, "Let bygones be bygones, and let us start in on a new basis, wiping out all old scores, and henceforth sticking to the true spirit of amateur sport!" The difficult problem is to find the individual or individuals who may have enough influence in both camps to bring about this greatly to be desired termination. This might possibly be accomplished through the alumni associations of the two schools at Harvard and at Yale.

Andover and Exeter men, after they get to college, usually become great friends, because they feel, as graduates of these large schools, a sort of superiority over their less-fortunate classmates who did not get their schooling in such great and well-known institutions. This gives them a common ground to meet upon, and they soon forget the petty differences they may have had before they became college-men, and their reminiscences become bonds of friendship rather than sources of disturbance. But if the alumni in the colleges are to do anything, they must first find out definitely from the men in the two schools that the body scholastic is willing to wipe away old scores and go into a new era of interscholastic contest, otherwise the same unfortunate comedy will be played that was enacted a year or so ago, when some graduates arranged for a reunion of athletic interests.

These arrangements could not have been very well laid out, for, at the school meetings held the same hour, the same day, at Exeter and Andover, different votes were reached. One school voted to resume friendly relations, and the other school, unfortunately, voted to maintain the same attitude



**REGINALD
FINCKE.**

**Interscholastic
Tennis champion
1896.**

that had caused the unpleasantness for the past two or three years. The school that held out the hand of friendship naturally felt hurt at this, and also naturally vowed that it would never take any steps toward making up again. But that time has passed, and let us hope those who were at the meeting have forgotten how they felt, and let us also hope that the other school has seen by this time that it is not well to refuse to shake the hand that is held out in good-fellowship.

So much is to be gained by both Exeter and Andover from the contests as they used to be ten or fifteen years ago that it would seem that both schools must yearn for the old order of things. Andover now seeks Lawrenceville as its rival, and although Lawrenceville is decidedly a strong opponent, and Andover-Lawrenceville games should never be discontinued, yet Lawrenceville is not Andover's natural rival. Exeter has sought Worcester Academy, and tries to make itself believe that it holds a great interest in the Exeter-Worcester games; but there is no doubt that there is not half the enthusiasm in the school over a Worcester victory as there used to be, and would be now, over an Andover victory. The same might justly be said concerning Andover and Lawrenceville.

It is for this reason that I so sincerely hope that the graduates will take some steps this fall, and open the new era of friendship with a football game. Let each school be willing to make concessions, and in the end they will find that they have gained tenfold in self-respect and self-satisfaction for each concession they made.

The seniors at these schools, and the men who graduated from them last year and are now entering Freshman year at college, can combine in this movement, and I feel sure that if they are careful and thoughtful in their methods, they can heal this breach which has been a painful one for every Exeter and Andover graduate, and an unexplainable one for outsiders who have looked upon these two schools as strong exponents of the healthy spirit of scholastic sport in America.



Beers. O'Rourke. Moore. Washburne. Hipple.

Bedford.

**THE N.Y.I.S.A.A. TEAM AT THE
NATIONAL GAMES.**

The new football rules for 1896 have at last appeared in book form. It is pleasant to know that they are a great improvement on anything of the kind we have ever had before in America. The thanks of all sportsmen are due to the University Athletic Club's committee for the work they have performed, and, as the season grows older, every football-player will realize more and more how much the reform of the code was needed.

There probably never was a more critical period in the history of football than last year, when there were three sets of rules, and when a certain number of colleges were playing under one set, an equally large number were playing under another set, and perhaps a larger number still were using the rules of the year before or a compromise between the factional codes.

The principal changes to be observed in the 1896 code are in the rules governing the fair catch and the scrimmage. The committee have thought it wise to bring back the fair catch to the old ruling, which requires that a mark be made with the heel; the old penalty of fifteen yards for holding has been retained. There were those in the committee on revision who thought that there should be a

penalty of twenty-five yards for holding a man who had made a fair catch, but the general opinion seemed to be that few umpires would have the courage to enforce this rule, and a compromise on fifteen yards was consequently adopted. Therefore, in so far as the fair catch is concerned, the situation is about the same as it was three years ago, except that the player making the catch is well protected by the severe penalty against interference or being thrown.

As to the scrimmage, the committee felt that there had been too great concentration of men in recent years, and they have attempted to hold in check the momentum plays. They have ruled that no player may take more than a single step before the ball is put in play—except one man, who may be in motion toward his own goal; this will bar out all forms of momentum play, and is a step in the right direction.

There is also a rule forbidding five men to bunch inside of tackles; this will serve to hold back mass plays, although, doubtless, the inventive minds of college football-players will be able to make up plays that can evade the spirit of this law. But the makers of the rules have hoped that the good judgment of captains and coaches may be relied upon to see that there is fully as much to be gained in open play as in the recently developed concentrated push work.

Among the minor changes to be noticed, in a cursory glance through the rules, is that on a fair catch the opponents must retire ten yards from the mark made by the catcher; this is a good ruling, for it places some value upon a fair catch. It is pleasant to note that the rule concerning interference with the centre rushers' snapping back of the ball has been made more stringent, and the officials must see that the ball is fairly put in play, and they must, according to the rule, insist that the opposing team do not interfere either with the ball or the man.

A number of years ago it used to be a favorite trick of the opposing centre to do all he could, by kicking or toeing or fingering the ball, to annoy the man who was snapping back. This was then put a stop to, and the opposing centres and guards began to shove and jostle their opponents, much to the delay of the game. These new rules concerning interference with centre play will prevent all of this nonsense, and will tend toward making scrimmage play more rapid and snappy.

On account of the development of quarter-back kicking, it has been found necessary to establish some rule that would make clear what kind of a kick must be made to give the opponents a fair chance at the ball. The rule has it that the ball when kicked by the quarter-back must pass beyond the line of the scrimmage.

One of the most complete and noticeable changes in the rules, however, although it is one that does not affect the game itself, is the common-sense arrangement of the paragraphs. The code begins by stating that "the game shall be played upon a rectangular field," etc. The old rules began with the statement that "a drop kick is made by letting the ball fall from the hands, and kicking it at the very instant it rises from the ground."

There is nothing in the old rules to show on what kind of a field the game should be played, or what the game was, or what kind of a ball it should be played with. There was nothing in the old rules forbidding a man to play the game with a cocked hat or a rubber boot if he chose; and if a team had come on a field with a baseball, and had insisted on running with that, there was absolutely no provision in the old rules by which the referee could forbid the playing of the game with a baseball.

Now, however, this absurd defect has been entirely done away with, and the new regulations, after describing the field and the teams, state that "the football used shall be of heavy leather, enclosing an inflated rubber bladder, and the ball shall have the shape of a prolate spheroid." Furthermore, no term is used in the new rules which has not first been fully defined. The first part of the code,

therefore, is made up largely of definitions, and this is of great assistance, for it will prevent many a discussion and dispute on the field.

A few more remarks in conclusion of what was said in the Department last week concerning preliminary football work: Remember that by practice alone can a team perfect itself; and consequently every manager should try to arrange as many practice games as possible for the eleven under his care. After the school season has fairly begun, and the neighboring schools have their elevens somewhat organized, it is always possible to arrange games with outsiders. Thus when the championship contests come along the players will be accustomed to games with strangers. Too much practice with a scrub team breeds slack play and indifference. If it is possible to schedule practice games with other schools for Wednesdays and Saturdays—playing some teams two or three times, perhaps, in one season—the manager should make such arrangements. In or near large cities this is almost always possible, and perhaps that is why the football elevens of New York and Boston are usually so well trained and developed.

The reason is very clear. When you play a team from a distance, you are bound to learn something of that team's methods, and it usually has some points which had not yet occurred to you. If you play that same team again two or three weeks later, it has learned from others, and puts its newly acquired knowledge in practice against you, and you again have the advantage of the other fellows' work. Of course your opponents are benefited in the same way through playing with you; but that is only right, and is much to be desired.

If it is possible for a school team to have the services of a coach—some graduate or some ex-football-player living in the neighborhood—it should avail itself greedily of the privilege, for it is too much to expect a young captain to handle his men well and do all the coaching besides. And when an older player has been invited to coach, his commands should be obeyed to the letter; for if the players had enough confidence in him in the beginning to desire or accept his services, they can do no less than carry out his instructions if he gives up his time to coach them in their sport.

Furthermore, it should be remembered that the coach has had greater experience than any of the players, and he can also—as an outsider—tell much better what the proper course of action for a team is than any member of that team, who may be influenced by a great many conditions that do not affect the coach, and so do not weaken his judgment.

"TRACK ATHLETICS IN DETAIL."—ILLUSTRATED.—8vo, CLOTH, ORNAMENTAL, \$1.25.

GRADUATE.

THE



This Department is conducted in the interest of stamp and coin collectors, and the Editor will be pleased to answer any question on these subjects so far as possible. Correspondents should address Editor Stamp Department.

I was misled by the telegraphic report of the A.P.A. convention. The annual dues are \$1.80, not \$1.08, as mentioned in last week's paper.

One of the weaknesses of human nature is a desire to get something for nothing. Many stamp-collectors are constantly on the lookout for a "big bargain" in rare stamps. Now bargains of this kind are never offered by first-class dealers, as they know the real value of their wares, but unprincipled persons take advantage of this weakness of humanity, and many a new collector finds reason to repent of having bought such "bargains."

A French journal, *Le Collectionneur des Timbres-Poste*, in a late number gives an amusing instance. Baron de M—— purchased a collection of rare stamps for 25,000 francs, which would have been a big bargain if appearances had not been deceitful. With the assistance of a well-known collector, the Baron discovered his stamps to be a marvellous collection of counterfeit, fake, and patched-up stamps.

The following are some of the varieties of tricks practised by unscrupulous persons:

1. Ordinary perforated stamps with exceptionally wide margins have their perforations trimmed off, and such stamps are offered as rare unperforated stamps.
2. Ordinary perforated stamps with wide margins are re-perforated with the rare perforations. This is frequently done by means of an ordinary hand punch.
3. Where stamps are printed in the same color with slight changes in lettering, the rare varieties are made by piecing. For instance, the one-franc French Empire is made by taking the 80 centimes, dark carmine, with the bottom label from the one franc of the Republic.
4. Bicolored stamps with the centre reversed, which are extremely rare, are made by cutting out the centre and reversing it on another copy of the same stamp. For instance, the 1869 U.S. 15c., 24c., and 90c. have been made by this process.
5. By chemical means the color is changed. For instance, the 10r. blue of Brazil is changed into the 10r. black.
6. Stamps which have been cancelled by pen and ink have their cancellation marks removed by chemical means, and these stamps are then sold as unused.
7. Counterfeit cancellations are frequently made on genuine stamps which have been surcharged "reprint" or "specimen."
8. Counterfeit surcharges are extremely common. They can be made on an ordinary printing press.
9. False water-marks are sometimes made by printing the stamps with wood-cuts, using a certain kind of oil, or they are made by pressing the design of the water-mark on the stamp, and then removing a portion of the paper by rubbing with pumice stone.
10. Very rare stamps of which a portion has disappeared have had these portions added.
11. Are the ordinary counterfeits. Sometimes these counterfeits are of a higher order of work than the originals. One of the great European houses made fac-simile copies of all the U.S. periodical stamps, and advertised the same as fac-similes. Each stamp bore the word "fac-simile" or "falsch."

These copies have frequently had a heavy cancellation applied to them immediately over the word "fac-simile" or "falsch." These stamps were then sold as genuine cancelled stamps.

The moral is a very simple one. Rare stamps should be bought from responsible dealers or responsible persons only.

MISS CECILE G. ROGERS, 118 B Bluff, Yokohama, Japan, wishes to exchange Japanese stamps for those of other countries.

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ADVERTISEMENTS.



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This Department is conducted in the interest of Bicyclers, and the Editor will be pleased to answer any question on the subject. Our maps and tours contain many valuable data kindly supplied from the official maps and road-books of the League of American Wheelmen. Recognizing the value of the work being done by the L.A.W., the Editor will be pleased to furnish subscribers with membership blanks and information so far as possible.

Several weeks ago we mentioned in this column another way of getting from Chicago to Waukesha. This was by taking steamer from Chicago to Milwaukee, and riding thence to Waukesha. The sail is a beautiful one, and makes one of the most interesting and varied pleasure trips near Chicago. On arrival in Milwaukee you will find yourself near the C. and N. W. Railroad depot, and the start awheel should be made from this point. Ride out Grand Avenue direct, crossing the river. This is a run of two and three-quarter miles, and when you are within about half a mile of the toll-house at the end of Grand Avenue, turn to the right and run up to it, keeping to the left on reaching it, and running thence out over the Watertown Plank-road. It is two and a half miles to Wauwatosa, and on entering the centre of the town turn left and cross the track. Thence run out up a steep hill, leaving Homewood on the south, and running into Elm Grove. On crossing the two tracks at Elm Grove, the road is clear to Brookfield, four and three-quarter miles away, and in good condition, except just as you leave Elm Grove, where there is a bit of hilly country. At Brookfield take the left hand of the three roads, and run thence direct to Waukesha. This three miles or more of road is in parts hilly, though at no point in the whole run is there any very bad hilly road. The road-bed itself is in very good condition for the whole sixteen or seventeen miles.

We have now covered this particular vicinity of Illinois pretty thoroughly. It is by far the best for bicycling on account not only of the good roads, but of the variety of scenery that presents itself to the wheelman. He can either ride by train from Chicago to Waukesha, or by boat and wheel, or by wheel. A very good week's trip would be to make the two days' bicycle trip from Chicago to Waukesha, as already described, stopping the night at Lippencott's. On the third day, or after a day's rest, to make the run to Oconomowoc and return to Waukesha, and then, with another day in the vicinity, to finally ride by wheel to Milwaukee over the road described this week, and return to Chicago from the latter place by boat. Or the trip could be reversed, and begun with the sail to Milwaukee, thence proceeding to Waukesha, and returning by the two days' trip to Chicago.

NOTE.—Map of New York city asphalted streets in No. 809. Map of route from New York to Tarrytown in No. 810. New York to Stamford, Connecticut, in No. 811. New York to Staten Island in No. 812. New Jersey from Hoboken to Pine Brook in No. 813. Brooklyn in No. 814. Brooklyn to Babylon in No. 815. Brooklyn to Northport in No. 816. Tarrytown to Poughkeepsie in No. 817. Poughkeepsie to Hudson in No. 818. Hudson to Albany in No. 819. Tottenville to Trenton in No. 820. Trenton to Philadelphia in No. 821. Philadelphia in No. 822. Philadelphia-Wissahickon Route in No. 823. Philadelphia to West Chester in No. 824. Philadelphia to Atlantic City—First Stage in No. 825; Second Stage in No. 826. Philadelphia to Vineland—First Stage in No. 827; Second Stage in No. 828. New York to Boston—Second Stage in No. 829; Third Stage in No. 830; Fourth Stage in No. 831; Fifth Stage in No. 832; Sixth Stage in No. 833. Boston to Concord in No. 834. Boston in No. 835. Boston to Gloucester in No. 836. Boston to Newburyport in No. 837. Boston to New Bedford in No. 838. Boston to South Framingham in No. 839. Boston to Nahant in No. 840. Boston to Lowell in No. 841. Boston to Nantasket Beach in No. 842. Boston Circuit Ride in No. 843. Philadelphia to Washington—First Stage in No. 844; Second Stage in No. 845; Third Stage in No. 846; Fourth Stage in No. 847; Fifth Stage in No. 848. City of Washington in No. 849. City of Albany in No. 854; Albany to Fonda in No. 855; Fonda to Utica in No. 856; Utica to Syracuse in No. 857; Syracuse to Lyons in No. 858; Lyons to Rochester in No. 859; Rochester to Batavia in No. 860; Batavia to Buffalo in No. 861; Poughkeepsie to Newtown in No. 864; Newtown to Hartford in No. 865; New Haven to Hartford in No. 866; Hartford to Springfield in No. 867; Hartford to Canaan in No. 868; Canaan to Pittsfield in No. 869; Hudson to Pittsfield in No. 870. City of Chicago in No. 874. Waukesha to Oconomowoc in No. 875; Chicago to Wheeling in No. 876; Wheeling to Lippencott's in No. 877; Lippencott's to Waukesha in No. 878.



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The sheriff of the village had been annoyed with complaints from the farmers about the loss of their chickens. He suspected a couple of colored gentlemen, and catching sight of them among the usual congregation at the village store one afternoon, he strolled in.

"Well, boys," he said, "there's a powerful rain-storm brewing in the west. I tell you that when that rain comes it will bring things above ground mighty lively."

In a short time the two colored men left the store. The sheriff chuckled to himself, and going after them, found them both busily burying chicken bones deeper in the earth.



Any questions in regard to photograph matters will be willingly answered by the Editor of this column, and we should be glad to hear from any of our club who can make helpful suggestions.

PRINTING ON COTTON, SILK, OR LINEN.

Photographic printing may be done on almost any textile fabric if it is properly sized and sensitized. Linen and silk may be obtained already sensitized, but it may be prepared very easily by the skilful amateur.

To size the fabric make a solution of

Boiling water	8 oz.
Chloride of ammonium	10 grs.
Iceland moss	6 grs.

When nearly cold filter through two thicknesses of blotting-paper, and soak the fabric in it for a quarter of an hour. Tack it at the corners to a smooth board so as to stretch out the wrinkles, and dry in a place free from dust.

Another sizing solution is made as follows:

Ammonium chloride 1 part
Water 125 parts
White of one egg.

After the fabric has been sized and dried it may be sensitized in any good sensitizing silver bath. A bath which is always reliable is made from

Nitrate of silver 60 grs.
Water 1 oz.

Dissolve the silver in the water, and add strong ammonia water drop by drop. A brownish precipitate will be formed, but keep on adding the ammonia till the liquid clears again. The mixture should be stirred all the while during the process, using a glass rod for the purpose. If the solution does not clear after the addition of twenty-five drops of ammonia, clear by filtering.

The fabric may be either immersed in this solution, or it may be stretched on a flat smooth board or sheet of glass, and the solution applied with a brush. The fabric should be stretched while drying, and if one uses the small-sized artist's thumb-tacks, there will be no danger of holes remaining in the fabric after it is taken from the board. It is almost needless to say that the sensitizing must be done in a room lighted by gas or lamp, and the fabric dried in the dark.

For printing one must make a special back for the printing-frame. For this take wood the same thickness of the back, and instead of making the two pieces which are hinged together of the same size, make one of them two inches, and the other six inches across—if the frame is a 5 by 8 printing-frame. A 5 by 8 printing-frame is a good size to use; the 4 by 5 is too small.

In the larger half of the printing-frame cut a round hole of two or three inches in diameter, and fit it with a smooth tapering cork. This cork should be of fine grain, and made to fit snugly into the hole.

To print, draw the fabric over the cork and push it up through the hole in the printing-frame far enough to come in close contact with the negative without pressing too hard against it. Fasten the springs and print as if on sensitive paper. Any boy or girl who is handy with tools can make one of these backs for a printing-frame.

One can make prints on the corners of silk handkerchiefs, on silks for cushion covers, on linen for doilies, or on linen for photograph-frames. The process is simple and inexpensive.

SIR KNIGHT RALPH B. ROOD, Cleveland, Ohio, asks if the radial energy of light is greater when the snow is on the ground than on clear days in summer; how dust can be kept from collecting on films during the drying; what makes the corners of a negative come out in some pictures with no detail, making heavy shadows

when primed. The light is much stronger in the winter when the sun is shining on the snow than it is in a clear day in summer, owing to the intense white glare reflected from the snow. A shorter exposure should be made for snow pictures, and a small diaphragm used. To keep dust from films when drying, pin the films by the corners to a flat board—the editor uses grape-basket covers for this purpose—and set the board in a slanting position against the wall, with films on the under side. The reason of the defective corners in the negative may be either from the film not being evenly coated, or from a shadow falling across the lens during the exposure. Sir Ralph has a No. 1 Kodak for sale, which is in good repair, and has a leather carrying-case. Sir Ralph's address is No. 23 Dunham Place, Cleveland, Ohio. Any member of the club wishing a Kodak cheap is requested to correspond with Sir Ralph.

SIR KNIGHT HUGO KRETSCHMAR, N. Y., asks if chloride of ammonia is the same as sal-ammoniac; if red prussiate of potash can be made of yellow prussiate potash; and if artists' gold-leaf could be used to make chloride of gold; and if an amateur could make nitrate of silver. Sal-ammoniac and chloride of ammonia are the same. Red prussiate of potash cannot be made from yellow prussiate of potash. If blue-print paper is made of yellow prussiate of potash (potassium *ferrocyanide*) instead of red prussiate of potash (potassium *ferricyanide*), the result will be a negative print, instead of a positive. The gold used by china-painters may be used for chloride of gold. Do not put the gold into *muriatic* acid, but into *nitro-muriatic* acid, made by mixing one part nitric and two parts muriatic acid, diluting with an equal quantity of water. This liquid is known as "aqua-regia." An amateur could make nitrate of silver, but it would not pay for the trouble, nitrate of silver being very cheap.

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About Order Chapters.

Here are two letters about Chapters:

I have read of many flourishing Chapters, and often wished I knew what was done to keep up the interest of the meetings. The Good Times Chapter, of Bridgeport, Conn., of which I was president, disbanded from lack of interest. We tried sewing while one read, entertainments by each member in turn, and a Chapter paper; but we tired of the first, gave up the second from lack of new ideas, and the third for the same reason and because the members did not wish to write for it. There is now some talk of reorganizing the Chapter, but unless something new is undertaken it will be useless. If members of Chapters, particularly those like ours, consisting of about half a dozen girls, would send some hints, I should be much obliged to them.

JOSEPHINE

BELDING.

12 WILLIAM

STREET, BRIDGEPORT, CONN.

The Columbian Chapter is in trouble. For many months it had largely attended meetings and jolly times. Then three members seemed bent on making trouble. We thought to disband and reorganize without them, but could not do so very well, because two of those proposed to be left out were our officers. There are six or seven new members to come in. What shall we do?

BRIGHT.

FOURTH AVENUE, DENVER, COL.

It is to be remembered that societies are aimed to effect improvement in their times, and to afford members a good time. Do not undertake to start a club and then find something to interest it to keep it in existence. Reverse the order. Find something to do, and then organize for the specific purpose of doing it. Pledge each member to a certain task. If there is lack of interest or apparent discussion, disband. When the foregoing conditions can be complied with, reorganize—but not until they can. As for suggestions for tasks, will some who have had success with various lines of work write to these inquirers? Order Chapters have been the sources of much profitable recreation, but always in cases where they set out to do certain things, and stopped when those things were accomplished or when interest flagged. An invaluable adjunct to a Chapter is an intelligent middle-aged lady who can and will make suggestions, settle disputes, etc. Those Chapters have been most successful that have had such help.

Kinks.

No. 28.—A MUSICAL MELANGE.

One starlit night in the month of roses I attended what was set forth on my invitation card of bay leaf to be a "Musical Melange." Though due at the Mountebank Club at nine, I was loath to miss this assuredly delightful midnight entertainment. A Brownie troop mounted on dormice hurried me out of the great bustling city to a grove of lindens and larches. In a cleared space in the centre I was within an hour introduced to my favorite musical geniuses, whom I little supposed would ever again appear upon this terrestrial sphere. As I look back upon the event, I can still see the glittering eyes of the goblins and ghouls who covered the branches of the near-by trees, and hear the whir and buzz of myriad crickets and katydids sweetly blended with the strains of a hundred musical instruments. Among those seated around a log glowing with fox-fire were these well-known musicians:

A vegetable, threw, and a consonant¹; to hew, and a preposition²; a part of the body, and a French article reversed³; the English equivalent of a sign of the Zodiac⁴; the outer bark of trees, a preposition, and a personal pronoun⁵; a masculine name meaning "a twin"⁶; a part of a priest's robe, a preposition, and a

vowel⁷; a hook⁸; a rope for capturing cattle⁹; a member of a secret society¹⁰; to trifle¹¹; a murmuring noise, and a small room.¹²

These spectral celebrities were each playing some instrument of soft and mellow tone. I noted with wonder that many of the instruments were bestudded with flashing stones. Among the instruments were:

To trifle¹³; a small wood¹⁴; to channel¹⁵; a shining bead of black glass¹⁶; to proclaim¹⁷; a sack, and large casks¹⁸; an iron pot, and part of the ear¹⁹; a city of Scotland²⁰; a famous cape²¹; a bird of New Zealand²²; dwells upon, a vowel, and harmony²³.

C

LEMENT RONALDSON.

No. 29.—A DUODENARY.

In each of the sentences is concealed the name of a celebrated poet.

1. The scarab urns of the Egyptians are at least bizarre.
2. As we gazed, horror-struck, Maguire's cot tottered, then fell.
3. "What a dasher, Bert! Throw the thing away."
4. A strong will is both fortunate and unfortunate.
5. Said the Ash, "The Aspens erred; the Abele was moaning."
6. "Too-whit!" exclaimed the owlets.
7. On the stand were scattered sundry dental appliances.
8. "It will be hard to reach 'em," snarled the guide.
9. While they were fumbling their creeses the trap opened.
10. The snug old smithy stood near the mill-stream.
11. Gleason's unruly red cow perambulated around the dairy.
12. The bearded stranger from the South eyed Charcourt suspiciously.

Z

OE D. ACKE.

Answers to Kinks.

No. 23.

Third column, *Isaac Newton*. 1. Cringe. 2. Tassel. 3. Prance, 4. Stamen. 5. Tocsin. 6. Ponder. 7. Clench. 8. Powwow. 9. Totter. 10. Crower. 11. Tinsel.

No. 24.—1. Genet. 2. Edile. 3. Nidor. 4. Eloin. 5. Terns.

No. 25.—Blunderbuss.

No. 26. Piano-forte.

No. 27.

A Gay Young Scot set out one day for a Hunt. He was thoughtful enough, Prior to starting, to Stow(e) away a lunch of Lamb and Bacon, and some Porter bought from a Brewer. Being a Lover of fishing, he carried also a Steel(e) Hook tied to a Reed. He wore a Brown Spencer and a Gray Hood.

As he was a Longfellow, he made Swift progress, till he stumbled over some Shell(e)y Knolls, and so got an aching side (Akenside). "How it (Howitt) Burns!" he exclaimed in a Stern, Savage voice. "It is enough to anger a Pope or a Bishop. But what are Wordsworth in curing a Pain?" he asked, with a Grim(m) smile.

He made a fire to Cook his fish, and while they were Browning he went to a coal-ridge (Coleridge) to dig for ore, with the intention of showing it to a Goldsmith to see if Sterling coin (Coyne) could be made of it. He dug until the sound of a Horn and a Campbell recalled him home.

Questions and Answers.

Charles Wood writes: "Will you kindly let me know what steps to take to become a cabin-boy on one of the ocean steamships? Also please tell me what is the pay of a cabin-boy, and whether there is a chance of advancement." Apply to the agents of the line; also to pursers of ships in port. Such positions are sometimes found on steamers plying to South American ports. A young man went recently on a steamship bound for San Francisco, *viá* Cape Horn, expecting to be about a year. He was required to furnish his own outfit of mattress, sheets, etc, sea chest, and heavy clothing. His pay was to be \$8 a month, and he was promised instruction in the rudiments of seamanship. There are apprentices in the engineer's as well as the sailing-officer's branch of the service, and advancement is promised in both. The work is very difficult, and the places are not easy to secure.

A. D. T. asks the meaning of the astronomical symbols which adorn the numerous calendars about Christmas-time. Each dot and quirk represents some emblem.



the Sun, is a bossed circular shield;



Mercury, represents the winged god's serpentlike *caduceus*, or wand;



aptly imitates the looking-glass of Venus, while Mars's symbol,



brings to mind the war god's helmet and plume;



Jupiter's sign, is an eagle, while



that of Saturn, forms a scythe;



is at once seen to be Neptune's trident. The signs of the ascending and descending nodes,



and



immediately suggest a croquet arch, or an eye of the hook-and-eye combination, but are, in point of fact, the head and tail of a dragon.

"Can you give me any information regarding the 'Daughters of the Revolution?' I would like to know who the founder is, where the headquarters are, and what the requirements are. Is there a society of this character that a child may join?—Eleanor C. Gardner." There are two adult societies of almost identical names. One is "Daughters of American Revolution," and the other, "Daughters of the Revolution." The first-named was organized in Washington in October, 1890. It aims to perpetuate the memory and the spirit of all who helped to achieve American independence, to acquire and protect historical sites, to erect, where possible, monuments thereon, and to preserve records, relics, and the like of early patriots and their acts. The conditions of membership are very liberal, being simply proof of descent from an ancestor (male) who fought loyally for independence. The age requirement is eighteen years. The president of the organization is Mrs. Adlai E. Stevenson, and the secretary Mrs. Donald McLean, 180 Lenox Avenue, New York city.

The other society is an offshoot of the former, organized in 1893, and membership in it is much more restricted. The president is Mrs. Edward P. Steers, 2076 Fifth Avenue, New York. The society for children is "The Society of Children of the American Revolution." This has its headquarters in Boston, but there are State organizations. The local societies are called chapters, and the chapter in New York city has about one hundred members. One of its objects is to form libraries, prominence being given to books on national subjects. Professor John Fiske has prepared a list of books for young students of American history. Applicants for membership must, as in the adult societies, prove their descent from active participators in the war for independence. The president is Mrs. Margaret Lothrop, Concord, Mass., who will, without doubt, have further information mailed if applied to.

Leo Reh binder writes to say that he enjoyed reading about West Point and Annapolis, and adds: "I do not think they would suit me. Can you name some universities or colleges for a poor boy, the cost, and chances in life after graduating?" All universities and colleges are for poor boys, in the sense that all lend every aid they can to brains and ambition that chance to belong to those in limited financial circumstances. Tuition is \$40 to \$150, with an average of \$75, but in every college there are free scholarships. Apply to the dean for conditions governing them. In not a few colleges there is no tuition at all, as in Michigan and most State universities, to pupils whose parents are citizens of the State. At Lehigh, located at South Bethlehem, Pa., there is no tuition charged any one. Board is \$5 per week, but there are students who live on less. In many colleges tutoring is to be had—sometimes enough to pay one's entire college expenses. As for "chances in life after graduating," no special answer can be given to that question. A general answer is that others succeed, and what others do you can try to do.

One Great Man's Method.

It is interesting to get a peep at the source of power, whether that source be an electric motor, a steam-engine, or a man while gathering material for great popular addresses. Mr. Francis Wayland Glen, who years ago was a partner in a nursery establishment in Rochester, New York, tells of a visit made by Henry Ward Beecher to his nurseries in that city. Having been shown everything, Mr. Glen asked if the great preacher cared to meet an educated Scotch gardener who had had thirty years' experience in the care of greenhouses. Mr. Beecher replied that he did.

Mr. Craig, the Scotchman referred to, was in the potting-house engaged in mixing potting-earth. He was a most retiring man, and as the party came upon him he was much confused at the announcement of the name of the renowned preacher. The latter

stepped forward and grasped the gardener's hand, disregarding the fact that it was covered with potting-earth, and shook it so warmly and so unconventionally that Mr. Craig was at his ease in a moment. Then began a remarkable series of questions. Mr. Beecher asking them, and for fully an hour, the Scotchman answering them with the confidence of an expert. Mr. Glen continues:

"I stood by and watched the operation with wonder and admiration. Mr. Beecher was gathering food and storing it away to digest and assimilate and give out to his parishioners. He was so cordial with Mr. Craig that he relieved him from all embarrassment, and he gave forth his answers with freedom and pleasure and great clearness. Plant after plant was taken up in the greenhouses, and its habits discussed, as well as those of the fruits in pots in the orchard-house. The parting was as cordial as the reception, or more so. Mr. Craig realized that he was appreciated by the great preacher, and Mr. Beecher recognized the fact that he had been receiving knowledge from a well-trained horticulturist and florist. It was a lesson to me, then a young man, having just passed my majority, that I have never forgotten, and the picture of one of the greatest teachers of men sitting at the feet of a plain, unpretending, unassuming gardener as a pupil is one I shall never forget."

Autography.

The omnipresent autograph-hunter has passed through many fortunate and unfortunate experiences. His hobby is somewhat of a lottery, bringing him a cold rebuff, or mayhap a prompt enclosure of the coveted autograph. An English nobleman once requested Talleyrand's autograph. The author promised to send one in a few days, and, true to his word, this note arrived:

"DEAR SIR,—Will you oblige me with your company to dinner on Wednesday next, at eight o'clock? I have invited a number of exceedingly clever persons, and do not like to be the only fool among them."

Daniel O'Connell, on being urged to pen an autograph, sent the following message to the stranger:

"SIR,—Yours requesting my autograph is received. I have been so bothered with similar impertinences that I'll be blest if I send it.

"

Your ob'd't servant,

DANIEL O'CONNELL."

The Rev. Dr. Sprague, of Albany, a Presbyterian pastor, was the possessor of a superb collection of autographs. He once requested the autograph of Benjamin Franklin from an eminent professor. "Oh, you have one already," said the professor. "No matter," replied Dr. Sprague, "I want it for exchange. One Benny Franklin in Europe is *worth two kings!*" Miss Alcott was always most patient with the ardent collector. She was once visited by a large club of young men, each one of whom wanted her autograph, and she did not refuse. Longfellow always kept a packet of autographs in his pocket, in case of need. And what malevolent person announced that a certain celebrity kept a rubber stamp of his *fac-simile* handwriting in his writing desk?

L

IONEL R. LANDON.

M

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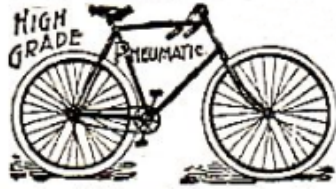
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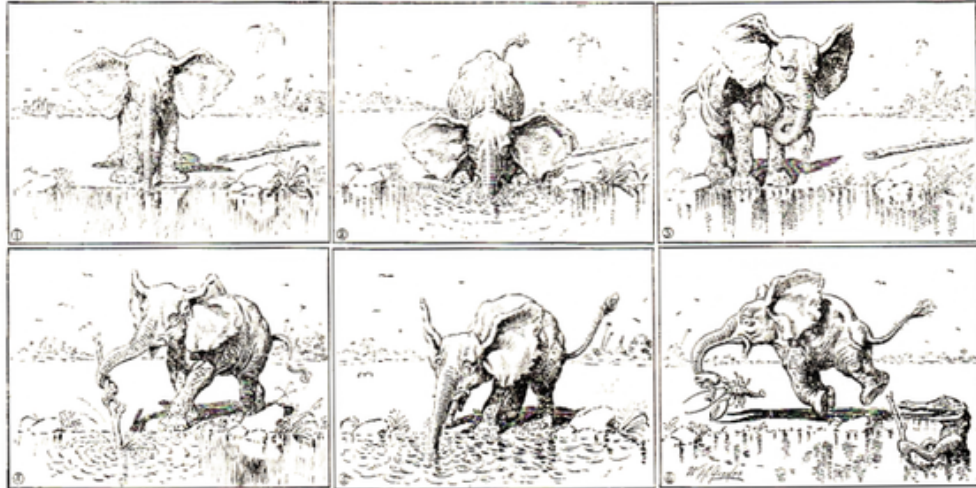
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HE STIRRED THINGS UP.

A MATTER OF STORIES.

"My uncle Jim," said Hal, "is building a house six stories high."

"That's nothin'," said Frankie. "My uncle George is writing a book ten stories long."

THE FIRST STEP.

"I suppose, Jacky," said the visitor to the little four-year-old, "that you will be going into business soon?"

"I'm going into pants first," said Jacky.

On one of the tram-cars that climb the hills in the streets of the little mountain town of D——, the driver is a quaint old man full of interesting and humorous characteristics. In the course of conversation with a passenger the other day he remarked that he and

his mule had been working steadily for the company for ten years, and that's a long time.

"Yes, it is," replied the passenger; "and surely the company must think a good deal of you both to keep you so long."

"Well, I've done honest work, and they know that, but I'm doubtful about how much they think of us. It was only the other day that the mule took sick, and the company got a doctor for the mule, and docked me for the time I lost. I dun'no', though; guess it was all right. Getting off here? Well, good-day, sir."

George Washington Jones, a colored gentleman, was sad, very sad. He was a kalsominer when he had work to do, but, as he expresses it,

"Dem dere white trash hab gone into de trade, an' now Ise got no work to do." But this was not what made him sad. "Dis yere life," he said, "am not wuth livin'."

"What's the matter, George," inquired his friend.

"Why, Ise got a little money on dat last job, an' Ise went round to settle de bills Ise owed."

"Didn't you attend to it all right?"

"Dat's de strange part of it. De butcher he wuz out, an' de grocer he wuz out, an' every one Ise wanted ter pay wuz out, an' den what'd I do but lose dat money."

"Well, that was unlucky, and no mistake; but still you showed your good intentions, and no doubt they won't press their claims."

"Press dere claims! Yah, dat's de trouble. When Ise got 'ome Ise found ebery one of dem waitin' to press dere claims, an' as Ise couldn't fix dem, dey done an' fixed me."

A STRIKING LIKENESS.

Miss Susan is an exceedingly refined young lady who has seen some five summers. She is full of airs and of graces, reserved, self-contained, and decidedly uppish. She cut her uncle dead in the street one day, and when he reproached her for her extreme hauteur, she said, with her most pronounced society manner,

"Oh, I saw you, uncle, but I thought it was auntie!"

There is an Irish porter employed in a large commission house in New York, one of the kind that will make a witty reply to any sort of question. He is very fond of expressing his views in general, and has great admiration of his arguments. If he fails to get a listener he will talk to himself in lieu of something better. A member of the firm being annoyed one day at his constant muttering, which he was unfortunate enough to hear, sent for him.

"See here, John, did it ever occur to you that your constant talk and muttering is a great annoyance to people that happen to be around? Why on earth do you chatter away to yourself, anyhow?"

"Shure I have two reasons fer doin' that."

"Two reasons! Well, what are they?"

"One of them is that I loike ter talk to a sinsible man, and the other is that I loike ter hear a sinsible man talk."

A RULE THAT DIDN'T WORK BOTH WAYS.

"What's your name?" said the new school-teacher, addressing the first boy on the bench.

"Jule Simpson," replied the lad.

"Not Jule—Julius," said the teacher. And addressing the next one, "What is your name?"

"Billious Simpson, I guess."

And the new teacher had to rap for order.

Teddy brought a green caterpillar in from the garden the other day, and showing it to his mother, he exclaimed, "I've got a big worm, mamma, but he ain't ripe yet!"

ALFRED'S DISCOVERY.

ALFRED. "This is a funny sort of an ice-cream-freezer, mamma."

MAMMA. "Why so, Alfred?"

ALFRED. "Because it freezes the ice-cream, and then goes and lets it melt."

BOBBY (*pointing to a fish jumping out of the water*). "Mamma, see that fish playing leap-frog!"

*** END OF THE PROJECT GUTENBERG EBOOK HARPER'S
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